

Hybrid Security Alarm System User Manual

August 15, 2023

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1. Introduction

The Hybrid Security Gateway is an IP-based multi-functional RF, optional Z-Wave, and/or ZigBee smart home security gateway with 4G/LTE and optional Wi-Fi capability, plus flexible hardwired and wireless sensor zone options, providing comprehensive solutions, including remote management, home security, live visual monitoring, home automation, environmental emergency monitoring, and energy management and being designed to bring all-round convenience, comfort and safety.

System Feature

- Ethernet / Wi-Fi / Cellular connection: Optional Wi-Fi connection provides backup to Ethernet to extend system flexibility.
- Multiple Communication Paths: The Hybrid Security Gateway has built-in RF, Optional Z-Wave, and/or ZigBee modules.

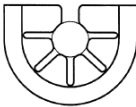
The Hybrid-Panel Series include the following models:

Name	Reporting Path	Modules
Hybrid-Panel-1-AC16/18V	IP/GPRS Reporting	IP Gateway supporting Sealed Lead-Acid (SLA) rechargeable battery (battery not included)
Hybrid-Panel-1-AC100/240V	IP/GPRS Reporting	IP Gateway supporting Sealed Lead-Acid (SLA) rechargeable battery (battery not included)

1.1 What's in the Box

Your package includes the following items:

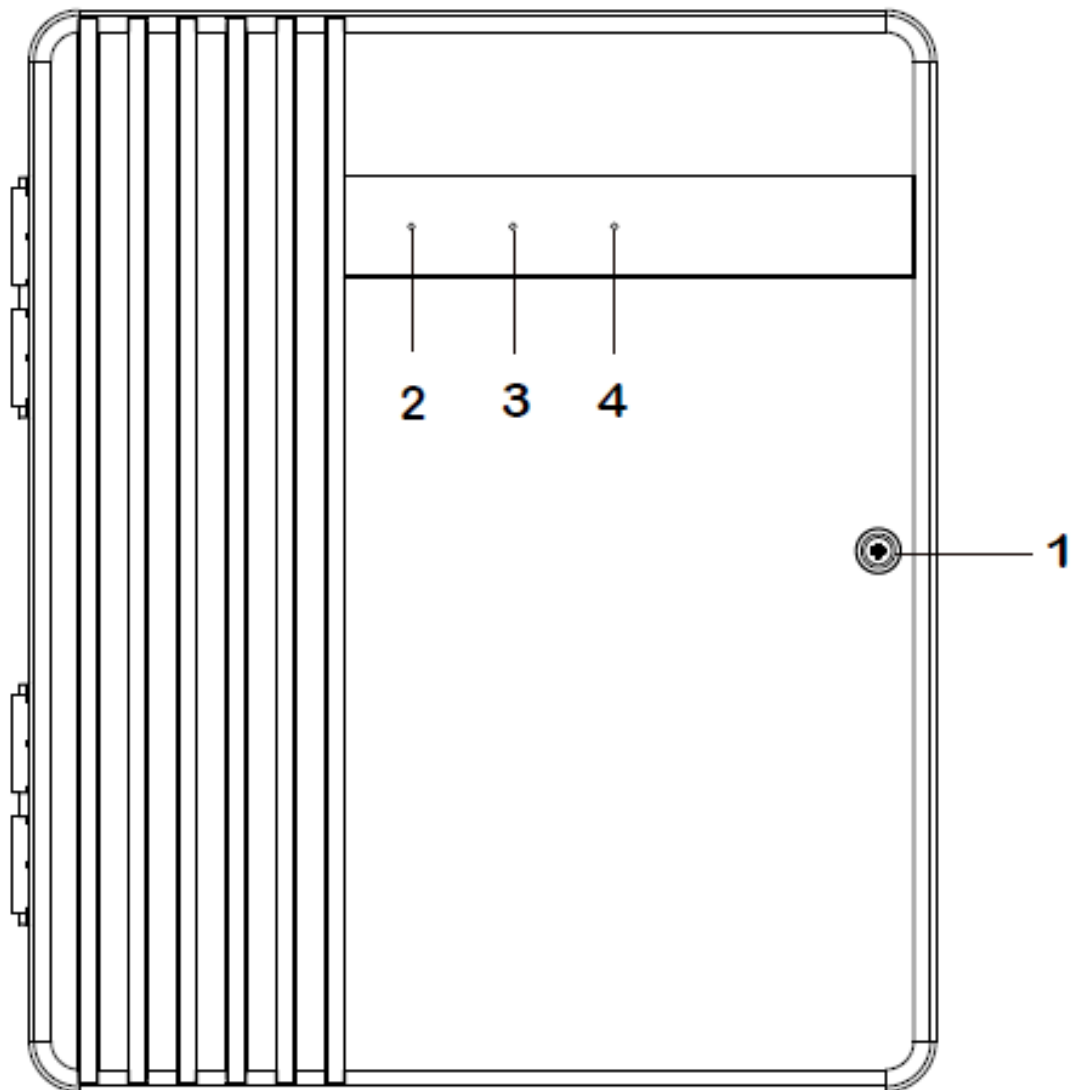
- Control Panel x 1
- Open Frame Power Cord x 1 (Hybrid-Panel-1-AC100/240V Only)
- An Ethernet Cable x 1
- Accessories:

Velcro tape x 4	
Jumper Connector x 2	
U-shaped Grommet x 4	
5.6K Resistor (2 resistors for each zone, 32 resistors for 16 zones in total)	

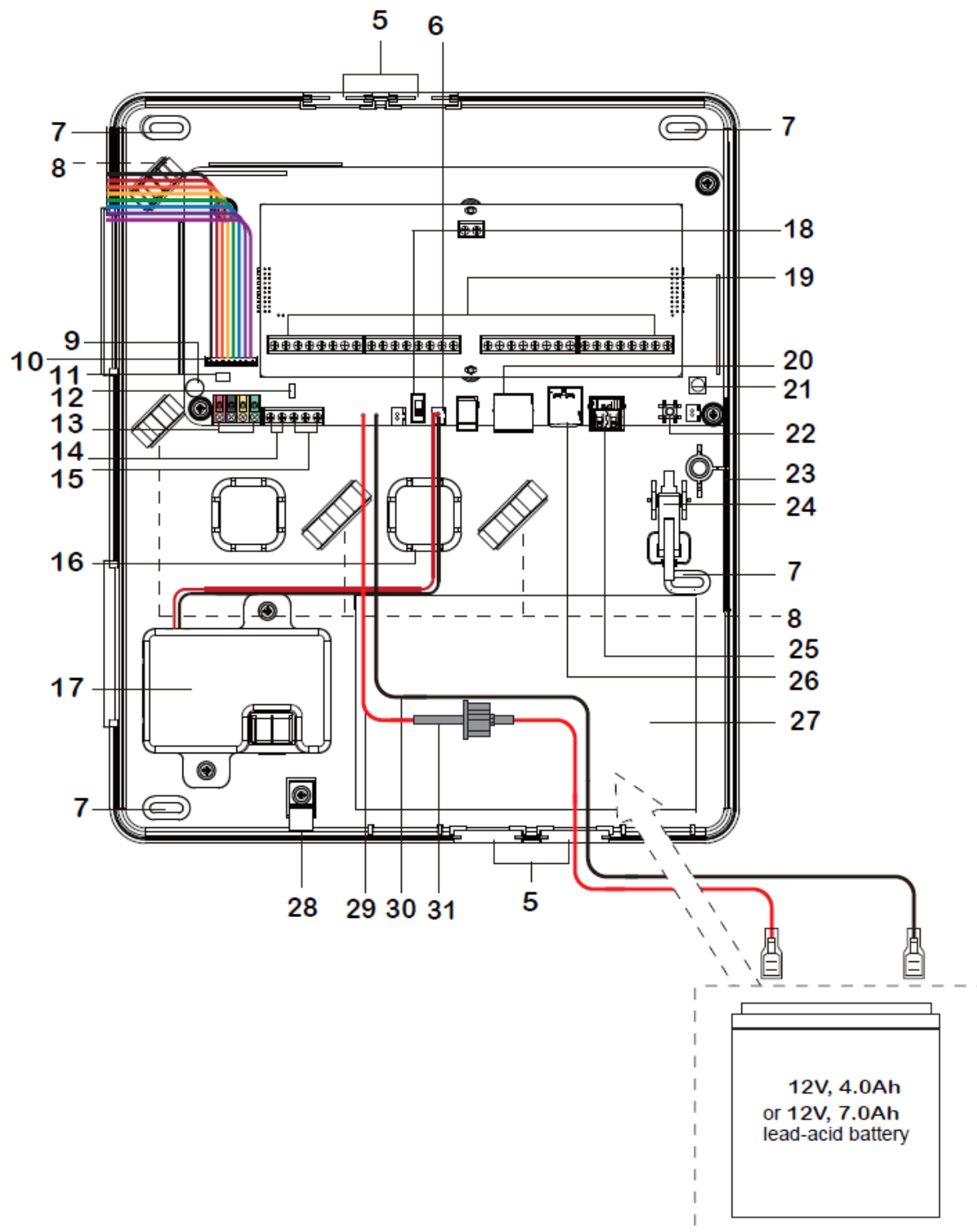
2. Panel Information

2.1. Identifying the parts

Top Cover (front view)



Back Cover (internal view)



1 Cover Fixing Screw

2 LED 1 - Area 1 (Green/Red)

- ☞ Full Arm mode - Red lighting up
- ☞ Home/1/2 mode - Red flashing
- ☞ Learning mode - Green lighting up
- ☞ Walk Test mode - Green flashing

3 LED 2 - Area 2 (Green/Red)

- ☞ Full Arm mode - Red lighting up
- ☞ Home/1/2 mode - Red flashing
- ☞ Learning mode - Green lighting up
- ☞ Walk Test mode - Green flashing

4 LED 3 - Status (Orange/Red)

- ☞ System Fault - Orange lighting up
- ☞ Alarm Trigger – Red flashing
- ☞ Alarm in Memory – Red lighting up

5 Removable Protective Cover x 4 (at top and bottom of the back cover)

Removable cover to reveal or hide cable wiring hole.

6 Built-in Power Unit Connector

7 Mounting Holes x 4

8 Wiring Clip x 4

To be used for securing power cables.

9 EGND Terminal

Please refer to **3.3 Hardware Installation** for detail.

10 LED Board Connector

The LED board connector is pre-installed onto the control panel.

11 J53 Jumper Switch

J53 Jumper Switch can be served as a terminating resistor, which can be turned to ON when wiring different devices to the panel to enhance connection.

12 J24 Jumper Switch

Please refer to **3.3 Hardware Installation** for detail.

13 Pluggable RS485 Terminal

Connect to a wired BUS device.

14 Bell Terminal

Connect to a wired bell box for bell wiring.

15 PGM Port

To be used as a voltage output port or a dry contact relay output depending on J24 jumper settings.

16 Alternative Hole for Wiring Management x 2

17 Built-in Power Unit

Input: 16 ~ 18VAC / 50VA ~ 70VA (refer to VA power selection table in **2.2 Power Supply**) or 100-240VAC.

AC module built-in power unit or Open Frame built-in power unit is installed. Use the built-in power unit to connect to the mains power.

☞ **Ensure to turn off all power supplies including Built-in Power Unit and SLA Battery before connecting or removing cables or wires.**

18 Battery Switch

Switch to “ON” for the battery to be charged when AC power is connected and serve as a backup power source when AC power is missing.

19 Zone Terminal & 13.5V Auxiliary Voltage Output Terminal & GND Terminal

Voltage output to zone 1-16 hardwired devices.

20 Ethernet Port

21 4G-LTE External Antenna Terminal

For connecting external antenna to the panel.

☞ **Do not disconnect the internal antenna when connecting the external antenna. To switch between the use of internal or external antenna, please configure through software setting. See chapter 7.1, GSM for details.**

22 Learn Button

To be used for local learning or local reset.

23 Internal Antenna PCBA

24 Tamper

25 USB Port (optional)

When connecting a USB dongle into the port, it is suggested to use a USB extension cable. There is no USB Port for the panel if it is built in 4G and Wi-Fi.

26 Micro-SIM Card Socket (not hot swappable)

27 Room for SLA Battery

Room for installing 12V 7Ah or 12V 4Ah rechargeable battery.

28 Wire Saddle

29 Embedded Battery Cable (Power) (Red)

30 Embedded Battery Cable (GND) (Black)

31 Tube Fuse Holder

Please note:

☞ **The tube fuse is pre-installed onto the battery cable; it is recommended not to change or replace the cable on your own to avoid possible hazards.**

2.2. Design Specs

Functions	
Number of partitions	2
Max number of wired/wireless zones	80 per partition, 160 total - Wireless and wired sensors, as well as the wired BUS devices, all count towards the total zone count. - Examples of maxed out sensor count: - o 160 wireless sensors learned in and 0 wired sensors - o 80 wireless sensors + 80 wired sensors - o 80 wires sensors + 40 wired sensors + 40 BUS devices
Number of available ports for wired sensors	- 16 on the Hybrid, capable of expanding up to 64 with expansion boards
Number of BUS devices	- 128
Number of PIR cameras (max limit)	16 (3-5 recommended)
Number of wired keypads (KP-35-COMBO)	4 (via RS-485 connection)
Number of users	40 total users with multi-partition access
User PIN Codes	1 PIN Code each user (6-digits, number 0~9), available combination from 000100~999999 (000000 and 000001 are disallowed). Note: Input of 5 invalid User PIN codes at the remote keypad will lock the remote keypad for 15 minutes.
Wired input	- NO/NC - Single end-of-line (SEOL) Double end-of-line (DEOL) - EOL resistance configuration (in ohms): 1K, 2.2K, 3.74K, 4.7K, 5.6K, 6.8K, 8.2K, 10K (32 pieces of 5.6K resistors are included in shipment)

Ethernet Interface	RJ-45 connection
Electrical	
External AC/DC adaptor	12VDC/2A
Built-in power unit	Input: variable per region - LATAM/APAC & MYR/SG/VNM/IL: 16~18VAC @40VA - o Dealer required to source 100-240VAC to 16.5VAC transformer - EMEA: 100~240VAC - JP: NA, Dealer required to source power supply Output: 12VDC/2A
Auxiliary output	12V/400mA (max.)
PGM output	12V/200mA (max.) Currently not supported
Backup Battery	Backup battery options: (1) 12V/7A Sealed Lead Acid Battery (can be expanded up to 2 pieces of batteries) (2) 12V/4A Sealed Lead Acid Battery (can be expanded up to 2 pieces of batteries)
Battery duration	(1) 12V/7A SLA: 36 hours (2) 12V/4A SLA: 21 hours
Wireless	
LTE Frequency	EU: B1 (2100 MHz) / B3 (1800 MHz) / B5 (850 MHz) / B7 (2600 MHz) / B8 (900 MHz) / B20 (800 MHz) US: B2 (1900 MHz) / B4 (1700MHz) / B12 (700MHz)
3G Frequency	EU: B1 (2100 MHz) / B5 (850 MHz) / B8 (900 MHz)

	US: B2 (1900 MHz) / B4 (1700MHz) / B5 (850 MHz)
RF Frequency	868.6375MHz / 433.82MHz / 426MHz / 919 MHz
Antenna type	On board (Z-Wave* / Wi-Fi / RF), External (4G Optional) *Only available on select Hybrid panels
RF encryption	Private Encryption Method
RF protocol	Climax F1
Z-Wave Frequency*	LATAM/APAC: 800 series Z-wave EU/US/JP: 800 series Z-wave *Only available on select Hybrid Panels
Wi-Fi	Dual-Band Cypress CYW4354 module 802.11 b/g/n, 2.4GHz
DECT Module	Reserved
Physical Properties	
Operating temperature	-10°C to 45°C (14°F to 113°F)
Operating humidity	85% relative humidity @23°C (non-condensing)
Dimensions	268 mm x 320 mm x 80 mm

2.3. Power Supply

Built-in Power Unit

- You can use the built-in power unit to connect to the mains power. AC module built-in power unit or Open Frame built-in power unit is installed. Please refer to ***Using the built-in Power Unit*** in **3.3 Hardware Installation** for more details.

Please note:

 **Ensure to turn off all power supplies including Built-in Power Unit and SLA Battery before connecting or removing cables or wires.**

Rechargeable Battery

- A rechargeable battery can be installed inside the Control Panel to serve as a backup in case of a power failure.
- During normal operation, AC power is used to supply power to the Control Panel and at the same time recharge the battery.

- If the battery switch is set as **OFF**, the battery will not be charged when AC power is connected and nor will it serve as a backup power source when AC power is missing. You need to switch the battery to **ON** for it to be charged when AC power is connected and serve as a backup power source when AC power is missing.

Power Output

- The panel supports up to a maximum total of 13.5V/2A (typical) for the current derived from VDD auxiliary voltage output terminals, Bell and PGM ports.
- The total current provided by Hybrid Panel for hardwired devices, bells, PGM, KPT-35-COMBO and expansion modules should not exceed 2A. Otherwise, additional power is required.

Please note:

 **If the total current exceeds 2A, components of Hybrid Panel could be damaged.**

VA PowerSelection Table

Sealed-Lead-Acid Battery Quality (pcs)	0	0	1	1
Output Current (A) including Vaux+Bell+Wired+PGM	1.5	2	1.5	2
Suggested 16VAC Input (VA)	50	70	70	90

Getting Started

Read this section of the manual to learn how to set up your Control Panel.

3.1. Selecting Mounting Location

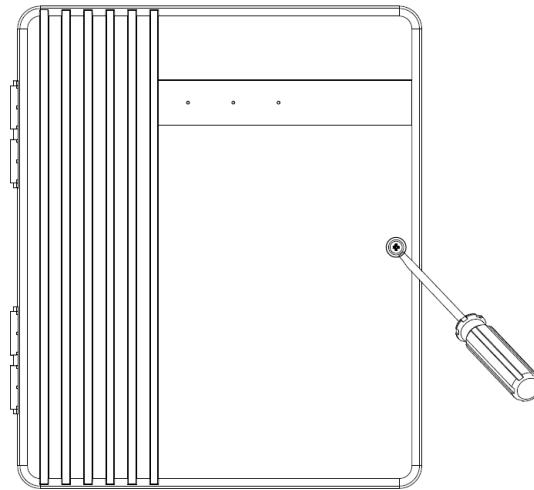
The Control Panel is designed to be wall mounted and protected against unauthorized case opening or removal from its mounting surface, follow guidelines below when planning installation location:

- The Control Panel requires the mains power and Ethernet connection.
- If a Cellular Module is used, ensure that there is good cellular coverage (Advisable to have a level of at least 4 out of 5).
- The Control Panel needs to have access to the routing of cables for the system to connect with wired devices.
- A central location between all the devices is often the best place, making wiring to expanders or devices easier, and preferably a place that is hidden from outside view.
- Avoid mounting the Control Panel in a damp location, close to a heat source, or near large metal objects, which may affect wireless radio strength.
- The Control Panel should be protected by sensors so that no intruder can reach the Control Panel without first activating a sensor.
- When using ZigBee/Z-Wave routers to improve ZigBee/Z-Wave network coverage, remember to use only ZigBee/Z-Wave Router with backup batteries for security sensors. If you use a Router without backup battery for security sensors, the Router will be powered down in case of AC failure, and your security sensors will lose connection with the ZigBee/Z-Wave network.

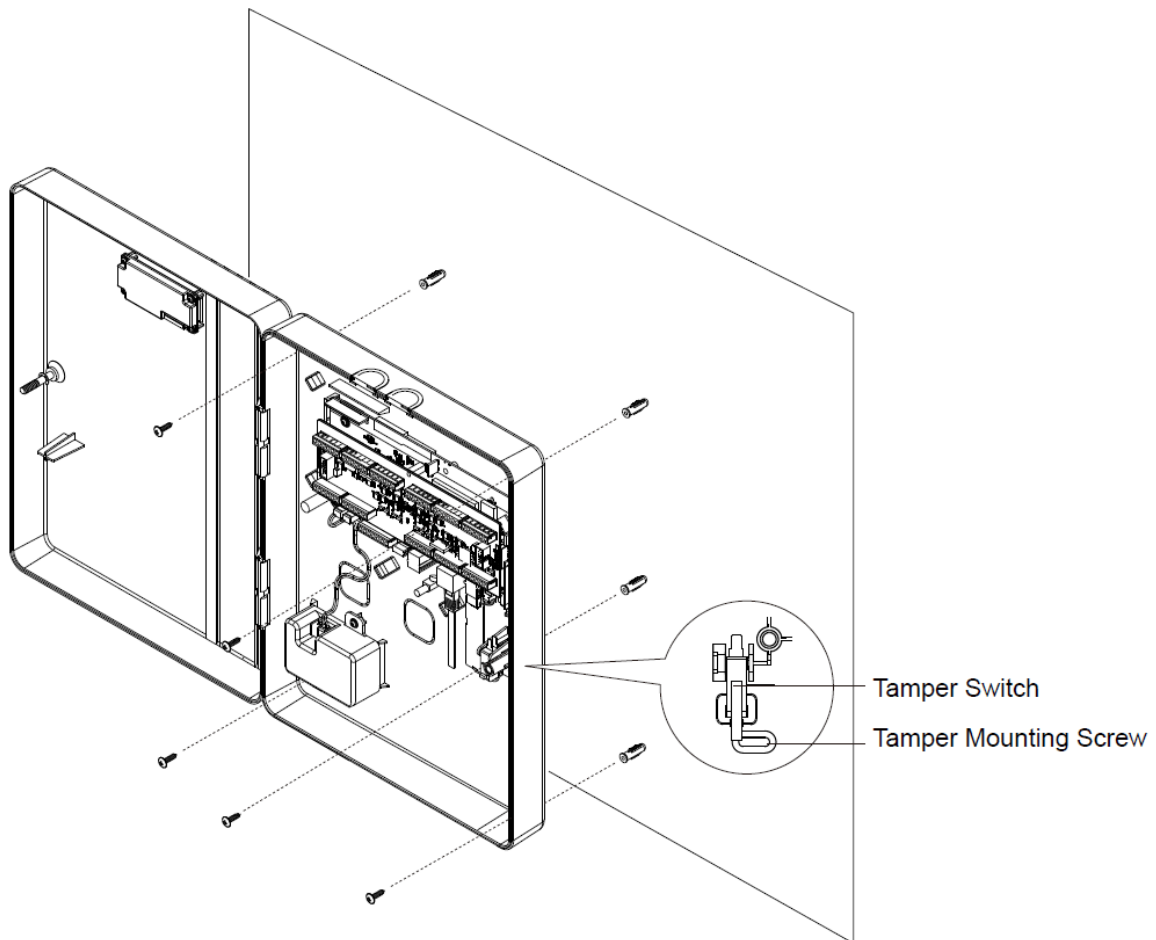
3.2. Mounting

Before installation or any maintenance work, make sure the power supply has been disconnected, and the battery switch has been slid to OFF position.

Step 1. Use a flat-head screwdriver to loosen the cover fixing screw to open the top cover.



Step 2. Use the 4 mounting holes as a template to mark and drill holes appropriately.



Step 3. Use the provided wall plugs for plaster/brick installation. Make sure the wall plugs are

flush with the wall.

Step 4. Connect the 4G-LTE external antenna to the antenna terminal on the panel.

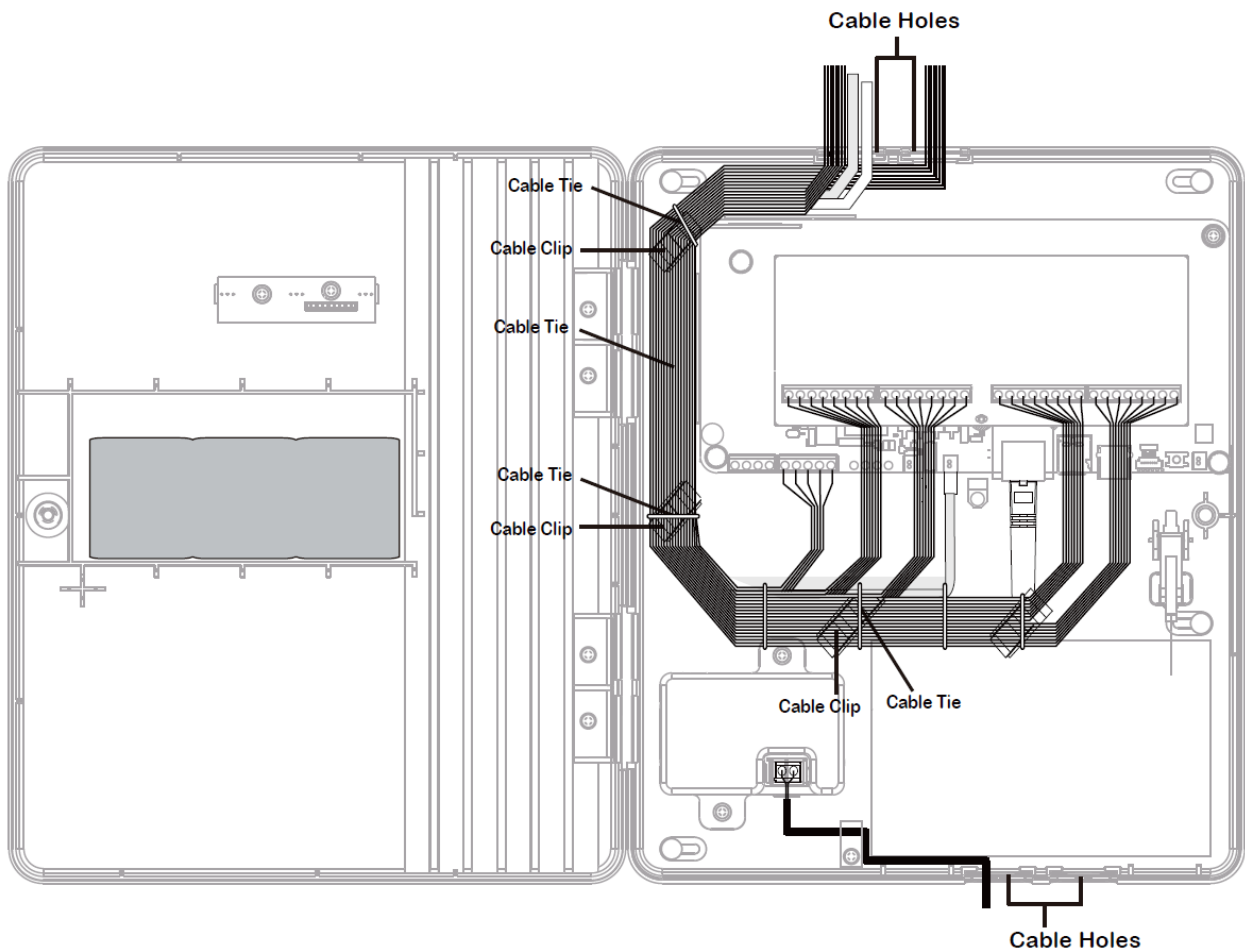
<NOTE>

- ☞ Make sure to connect the antenna when the Panel is powered off.
- ☞ Do not place the antenna close to metallic materials.

Step 4. Screw the Control Panel onto the wall. Make sure the Tamper Switch is fully depressed against the wall

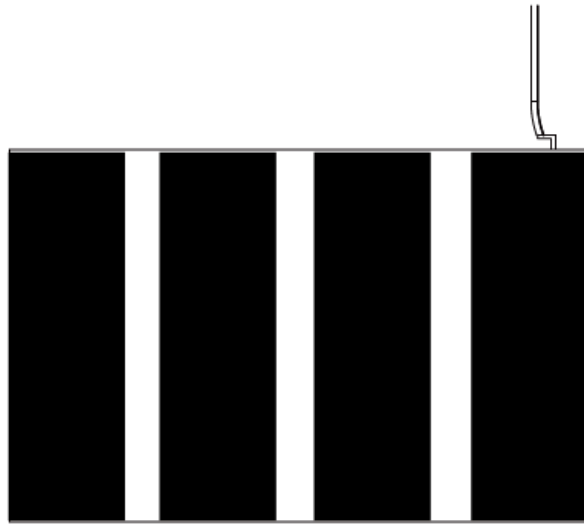
Step 5. Complete wiring following the instructions in later section **3.3. Hardware Installation.**

Step 6. Arrange all wires along the wire clips and then route the wires to the two cable holes on top or bottom. Use cable ties (not included) to thread through the wire clips and secure the wires. Use the U-shaped grommets around the cable holes to manage wires.



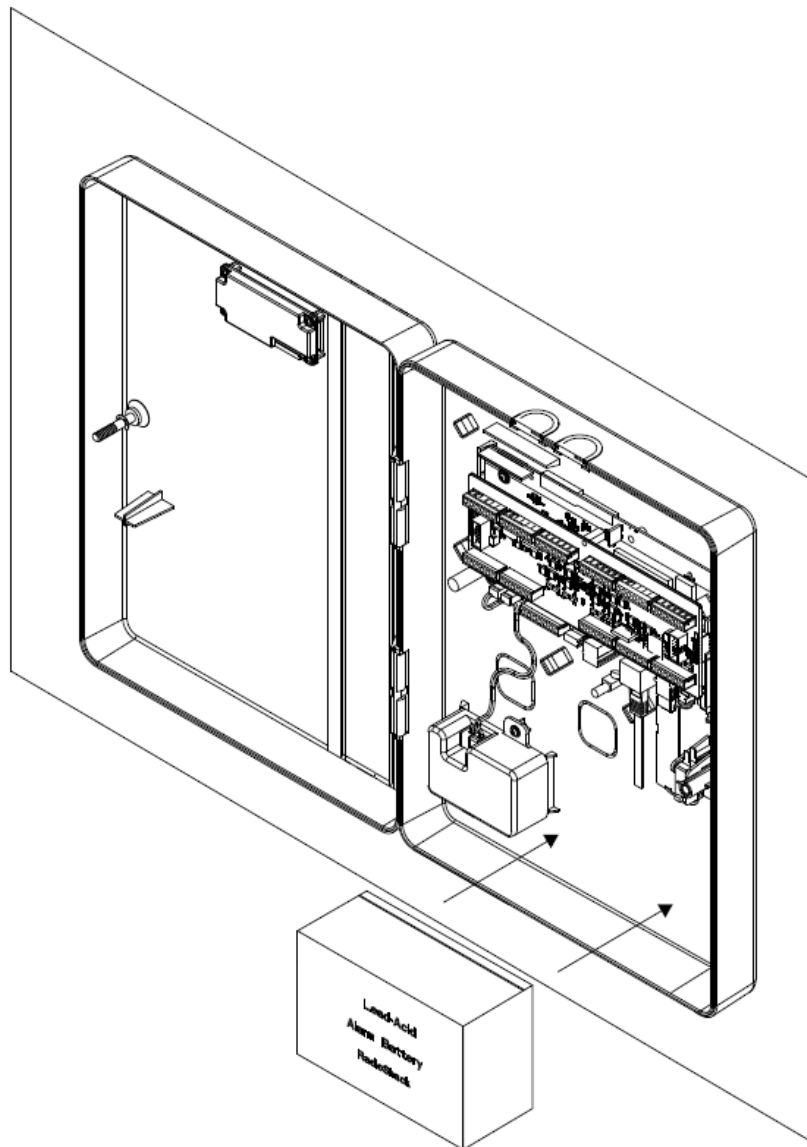
Step 7. Take out 4 Velcro tape(s) (Loop side) from the package.

Step 8. Peel the release film of the tapes. Apply them to your battery and press firmly for 30 seconds to ensure good contact. Please avoid applying the tape on uneven surface or re-apply it.



SLA Battery with 4 Velcro Tapes (Loop Side)

Step 9. Attach the battery to the hook side with Velcro Tape on the Hybrid Panel.



Step 10. Close the cover and tighten the cover fixing screw.

3.3. Hardware Installation

CAUTION:

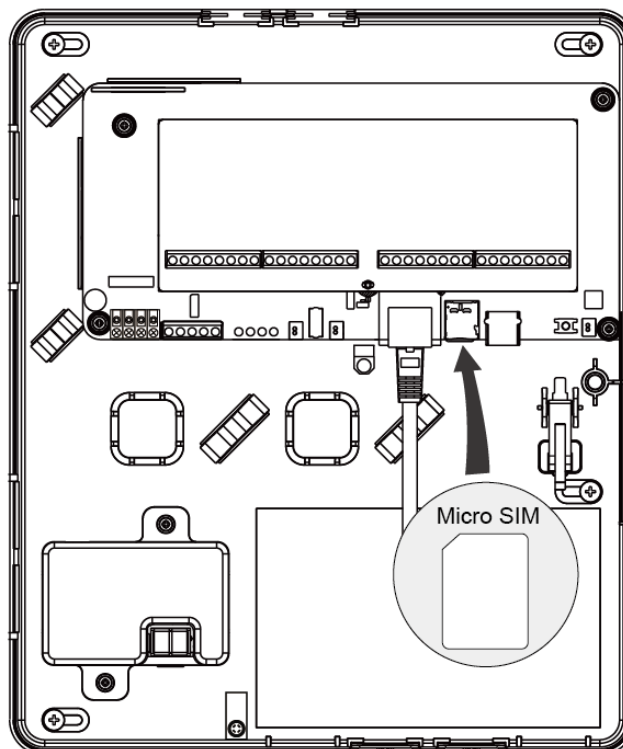
- **Before servicing, make sure the AC power has been disconnected, and the battery switch has been slid to OFF position.**
- Wiring of the Hybrid Alarm System should only be performed by certified technician with proper knowledge and training in electric equipment.
- Before installation or any maintenance work, make sure the power supply has been disconnected, and the battery switch has been slid to OFF position.
- Do not connect the devices to loads exceeding supported load current.
- Do not connect the battery or the built-in power unit until all wiring is complete.
- The control panel enclosure must be secured to the wall before operation.
- Internal wiring must be routed in a manner that **prevents**:
 - Wiring over circuit boards
 - Excessive strain on wire and on terminal connections
 - Loosening of terminal connections
 - Damage of conductor insulation
- Incorrect connections will result in failure or improper operation. Inspect wiring and ensure proper connections before applying power.

Step 1. For configuration and operation of the Control Panel via Ethernet, connect an Ethernet cable to RJ-45 port.

Step 2. (Optional) Insert a Micro-SIM card with the gold (metal) side facing down, as example shown below. Before inserting the SIM card, please make sure the pin code is deactivated and SMS messages are removed first.

<NOTE>

- ☞ Inserting the SIM Card when the Panel is powered off.
- ☞ Make sure to insert a SIM card with data plan.

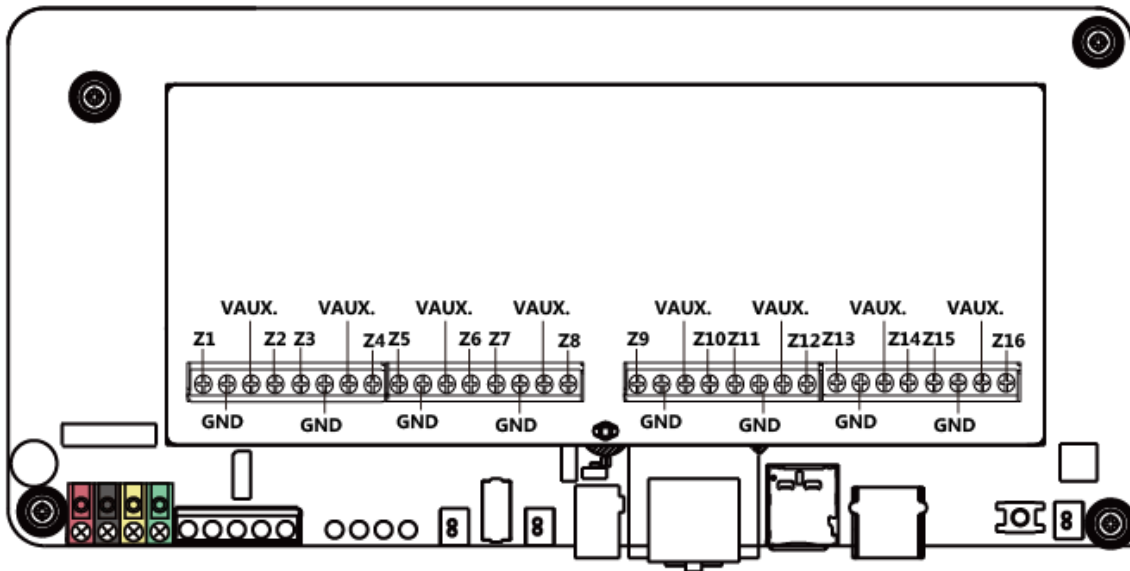


Step 3. Complete wiring for Zone 1-16 terminals, GND terminals, AUX power terminals, PGM

terminals, Bell terminals, and Keypad terminals. (Please see the following sections for details.)

- Step 4.** Connect the battery to the PCB board. See **Battery Installation** in the following section for more details.
- Step 5.** Connect to the mains power using the built-in power unit.
- Step 6.** Slide the battery switch to ON position.

Zone Wiring (Zone 1 -16)



- The 16 zones can be wired by supervising NC (normally close) or NO (normally open) devices, e.g. PIR sensor, door contact, smoke detector, water sensor, fire sensor, CO sensor, gas detector, heat detector, and glass break detector, etc.
- Wire gauge: Minimum 20 AWG, maximum 18 AWG. Do not use shielded wire.
- The hardwired zones support Single-End-of-Line (SEOL), Double-End-of-Line (DEOL) loop configuration, with selectable resistor values of 1K Ω , 2.2K Ω , 3.74K Ω , 4.7K Ω , 5.6K Ω , 6.8K Ω , 8.2K Ω , 10K Ω ohms.
- Triple end-of-line (TEOL) loop can be configured in different combinations: 4.7K Ω , 6.8K Ω , 12K Ω (resistor value selection: 6.8K), 4.7K Ω /5.6K Ω , 4.7K Ω , 2.2K Ω /3K Ω (resistor value selection: 4.7K), or 4.7K Ω /5.6K Ω , 5.6K Ω , 2.2K Ω /3K Ω (resistor value selection: 5.6K).
- For an NO loop, please have an EOL resistor in parallel (across) the loop.
- For an NC loop, please have an EOL resistor in series with the loop.
- If a zone wiring method is changed, be sure to turn the system power off and back on again to avoid triggering the alarm.

Please refer to the following diagrams of loop 1 to 10 for wiring examples.

NO/NC Wiring

The panel can detect alarm for corresponding NO or NC devices via the open, secure or shorted circuits.

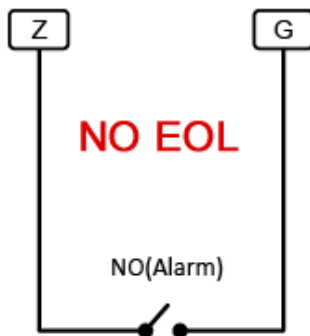
Total wiring length limit for connected NO/NC devices:

- Max. 3000 ft / 914 m @ AWG-22
- Max. 4900 ft / 1493 m @ AWG-20
- Max. 6200 ft / 1889 m @ AWG-19

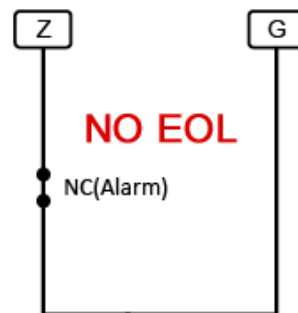
<NOTE>

- ☞ The wiring length is calculated based on maximum wiring resistance of 100Ω (50×2), which is multiplied by 2 because the wire is round-tripped. Please be noted that the wiring on any device must be a pair (two wires), with total resistance <100 ohms for dual measurement (<50 ohms for single wire).

1.



2.



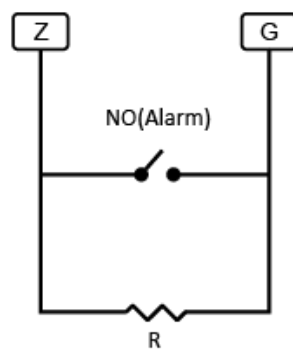
<NOTE>

- ☞ There is no EOL resistor in loop 1 and loop 2

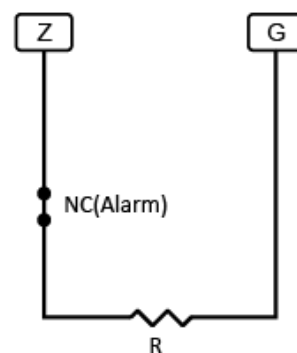
Single-End-of-Line (SEOL) Resistor Wiring

The single-end-of-line (SEOL) resistor shall be installed near the wired device at the end of a zone loop to supervise the wiring conditions for NO and NC devices, so the panel can detect alarm and tamper for corresponding devices via the open, secure or shorted circuits.

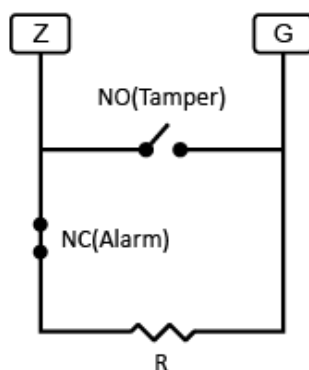
3.



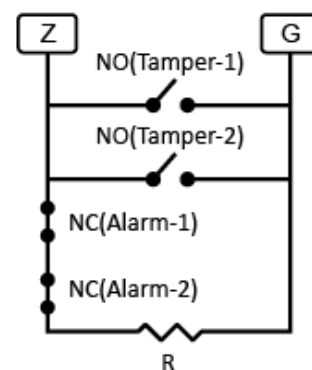
4.



5.



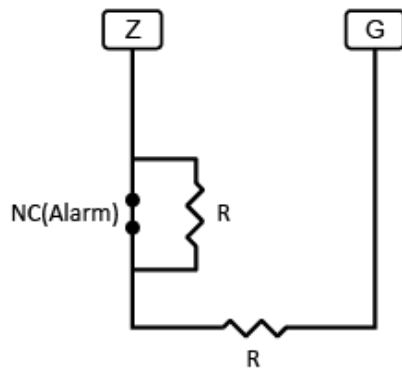
6.



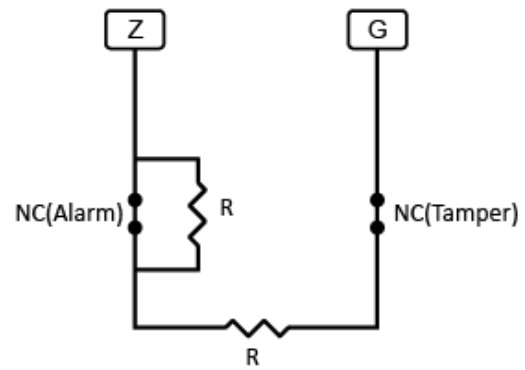
Double-End-of-Line (DEOL) Resistor Wiring

The double-end-of-line (DEOL) resistor shall be installed near the wired device at the end of a zone loop to supervise the wiring conditions for NC devices, so the panel can detect alarm and tamper for corresponding devices via the open, secure or shorted circuits.

7.



8.

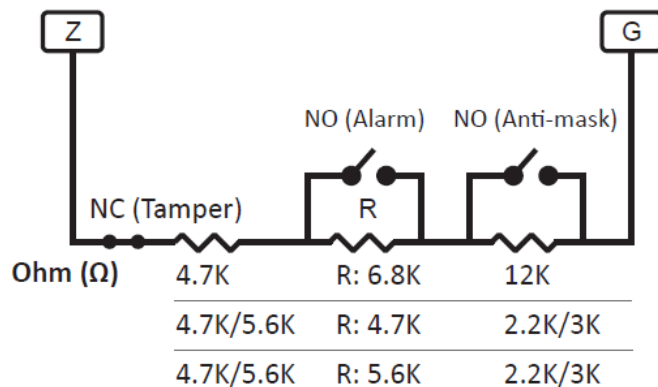


Triple-End-of-Line (TEOL) Resistor Wiring

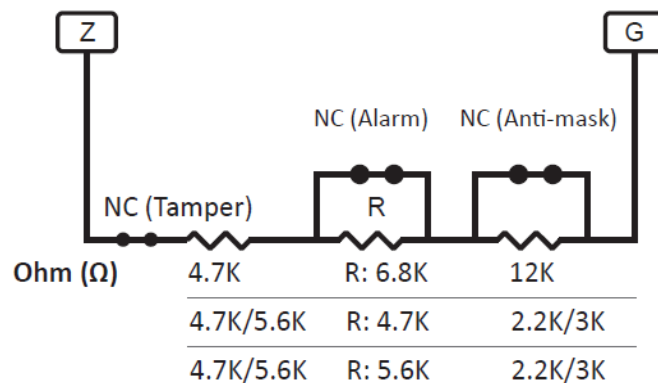
The triple-end-of-line (TEOL) resistor shall be installed near the wired device at the end of a zone loop to supervise the wiring conditions for NC or NO devices, so the panel can detect alarm, tamper and anti-masking for corresponding devices via the open, secure or shorted circuits.

☞ The unit for values in the following is in ohms (Ω).

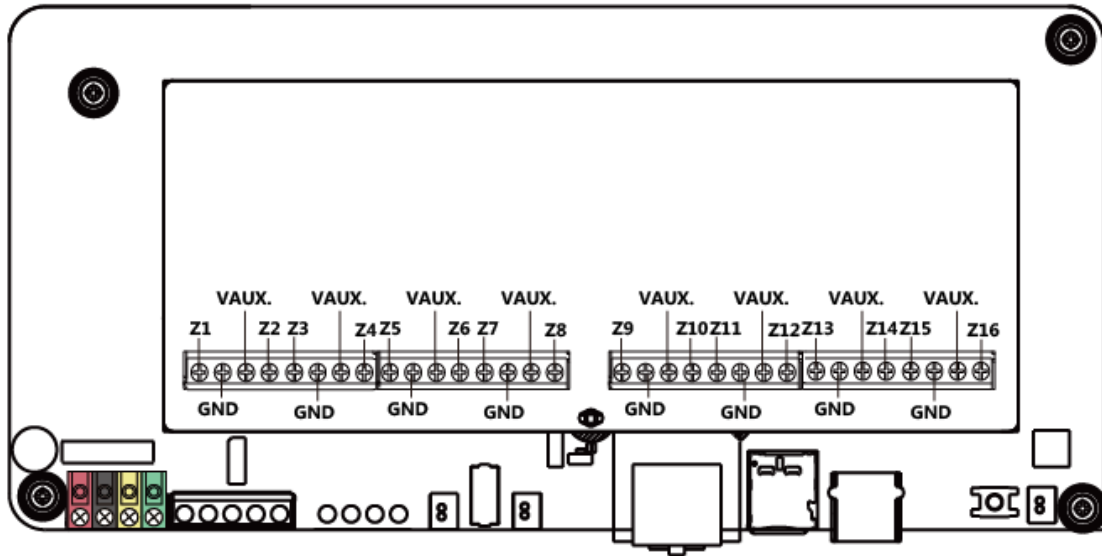
9.



10.

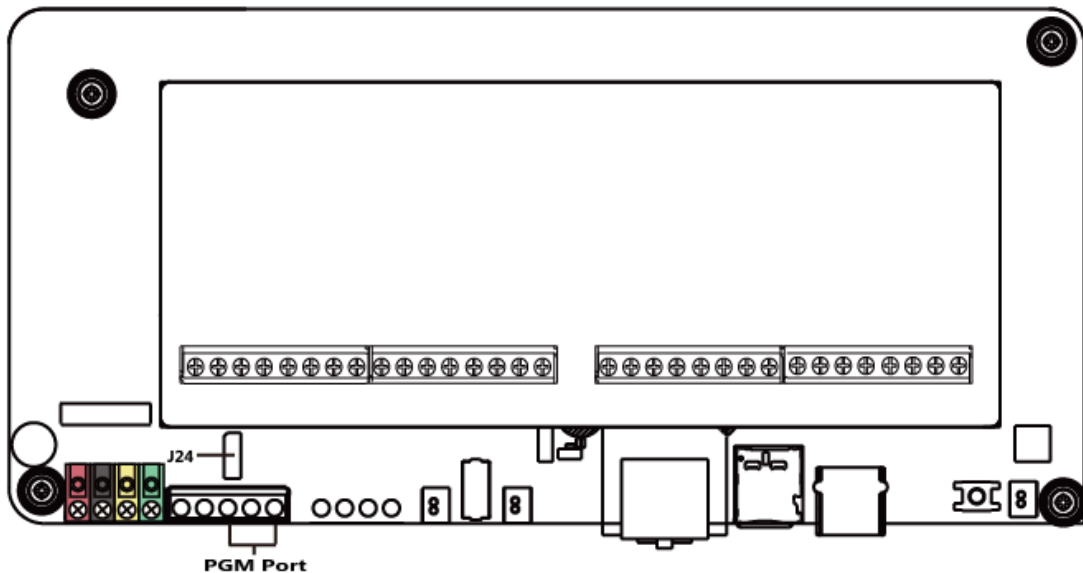


AUX Power Wiring

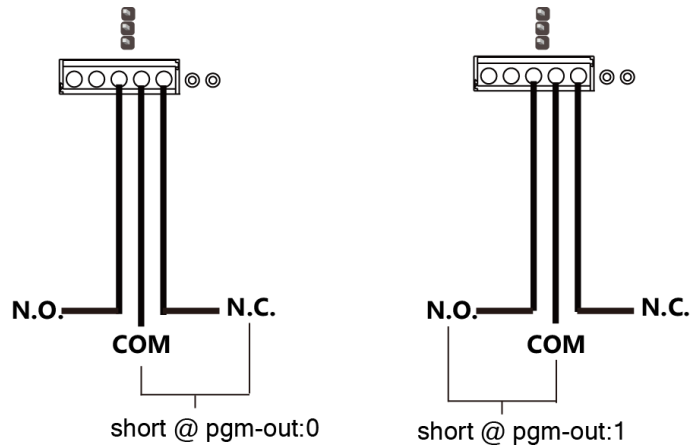


- The Control Panel can provide a maximum total of **2A** current for hardwired zones, auxiliary output, keypad (KPT-35-COMBO), wired expansion module (WEZC-8, WEZ Series), output expander (WEPC-1), and PGM/Bell ports.
- Min/Max operating voltages for devices/detectors is 10 VDC -14VDC.
- Please note that the total current should **not exceed 2A**, otherwise, an additional power supply is required.

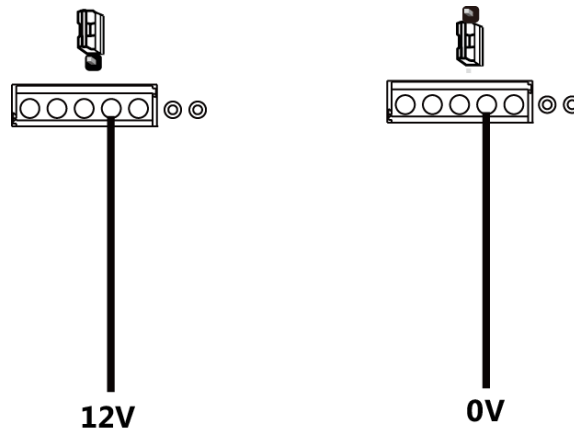
PGM Wiring



- The output of PGM is 12V voltage with a maximum current of 500mA.
- The port can be a voltage output port or a dry contact relay output depending on J24 jumper settings.
- When J24 Jumper Switch is disconnected, PGM port will operate as a dry contact relay output.
- Please be sure to assess the specifications of the connected devices to deduce the appropriate wiring for the PGM connection.

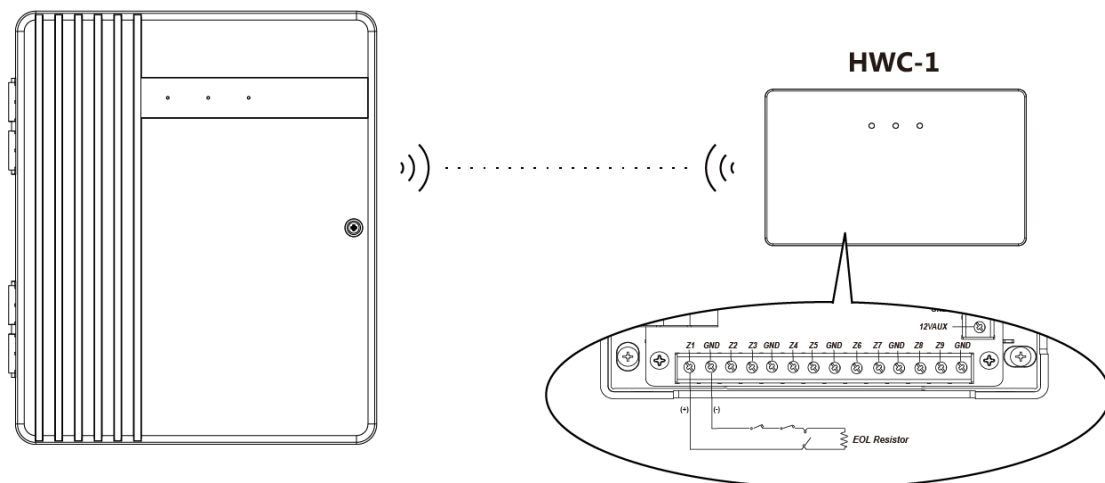


- When the J24 jumper link is inserted connecting the 1st and 2nd pins, the PGM port will provide 12V output.
- When the J24 jumper link is inserted connecting the 2nd and 3rd pins, the PGM port will provide 0V output.
- Min/Max operating voltages for devices / detectors is 10VDC -14VDC.
- The total current derived from auxiliary output, bells, and PGM ports should not exceed **2Ah**.



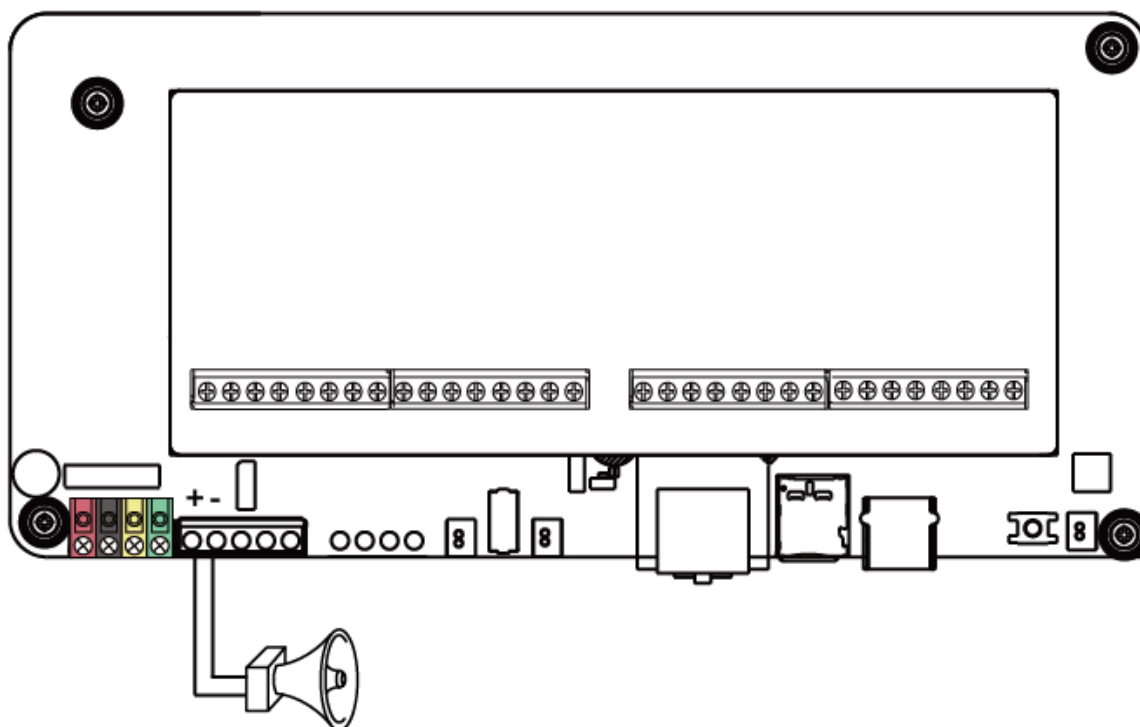
Wired Zone Expansion

- Hybrid Security Alarm System is compatible with HWC-1 Wireless Converter. One HWC-1 can add 9 wired zones to the Control Panel.



Bell Wiring

- Be sure to observe polarity when connecting polarized sirens or bells.
- Min/Max operating voltages for devices / detectors is 10VDC -14VDC.
- The total current derived from auxiliary output, bells, and PGM ports should not exceed **2A**.



Connecting Keypads / Wired Security Devices / Expansion Modules via the data BUS

The keypads, security devices, and expansion modules compatible with Hybrid Panel can be connected in series via the data BUS.

To assist with cable connections, the terminal blocks on each system module are color-coded for easy identification.

Terminal block color codes:

Red	VDD
Black	GND
Yellow	485A
Green	485B

Note:

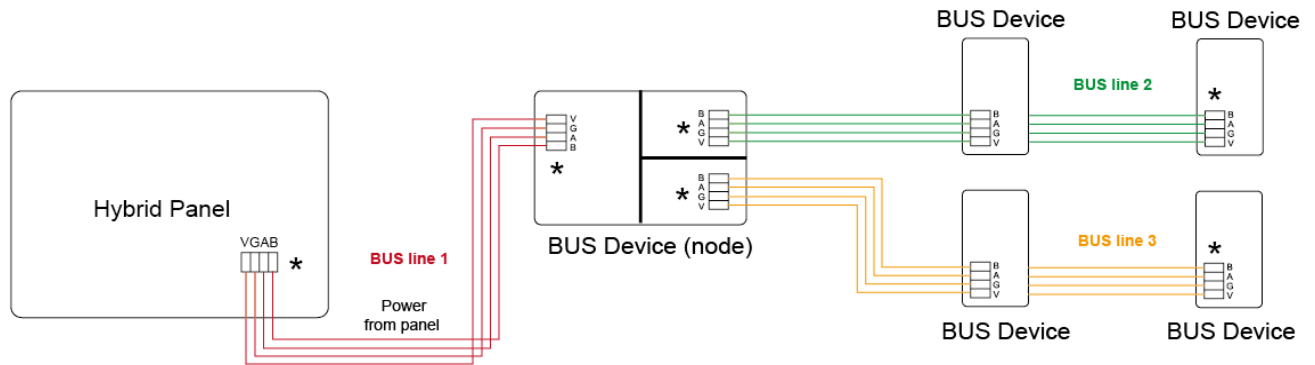
For KPT-35-COMBO and KPT-35(N)-BUS, the terminal blocks are marked as V, G, A, B instead of color-coded.

Wiring Guideline

- The Hybrid Panel is the first device on the data BUS. Ensure the Control Panel's J53 Jumper Switch and the furthest BUS device's Jumper Switch are set to ON to serve as a terminating resistor.
 - ☞ Be sure to only enable the aforementioned 2 jumper switches and do not set the jumper switches to ON for any other BUS devices in between.
- The total wiring length limit is a maximum of 3000 ft / 914 m.
 - ☞ Depending on wired device's power consumption, PWB-1-BUS (auxiliary power supply module) or AMP-1-BUS (amplifier) might be needed.
- The total number of BUS devices (refer to as "nodes"; the Hybrid Panel is counted as one node) on each BUS line must be within 32 or less. Otherwise, BUS signal abnormalities may occur. Be noted that a maximum of 128 BUS devices (including Hybrid Panel and other

nodes) can be connected to the panel.

- In the example below, there are a total of 3 BUS lines/segment: **BUS line 1** contains 2 nodes (Hybrid Panel and ISL-BUS), **BUS line 2** contains 3 nodes (ISL-BUS and 2 BUS devices), **BUS line 3** has 3 nodes (ISL-BUS and 2 BUS devices).



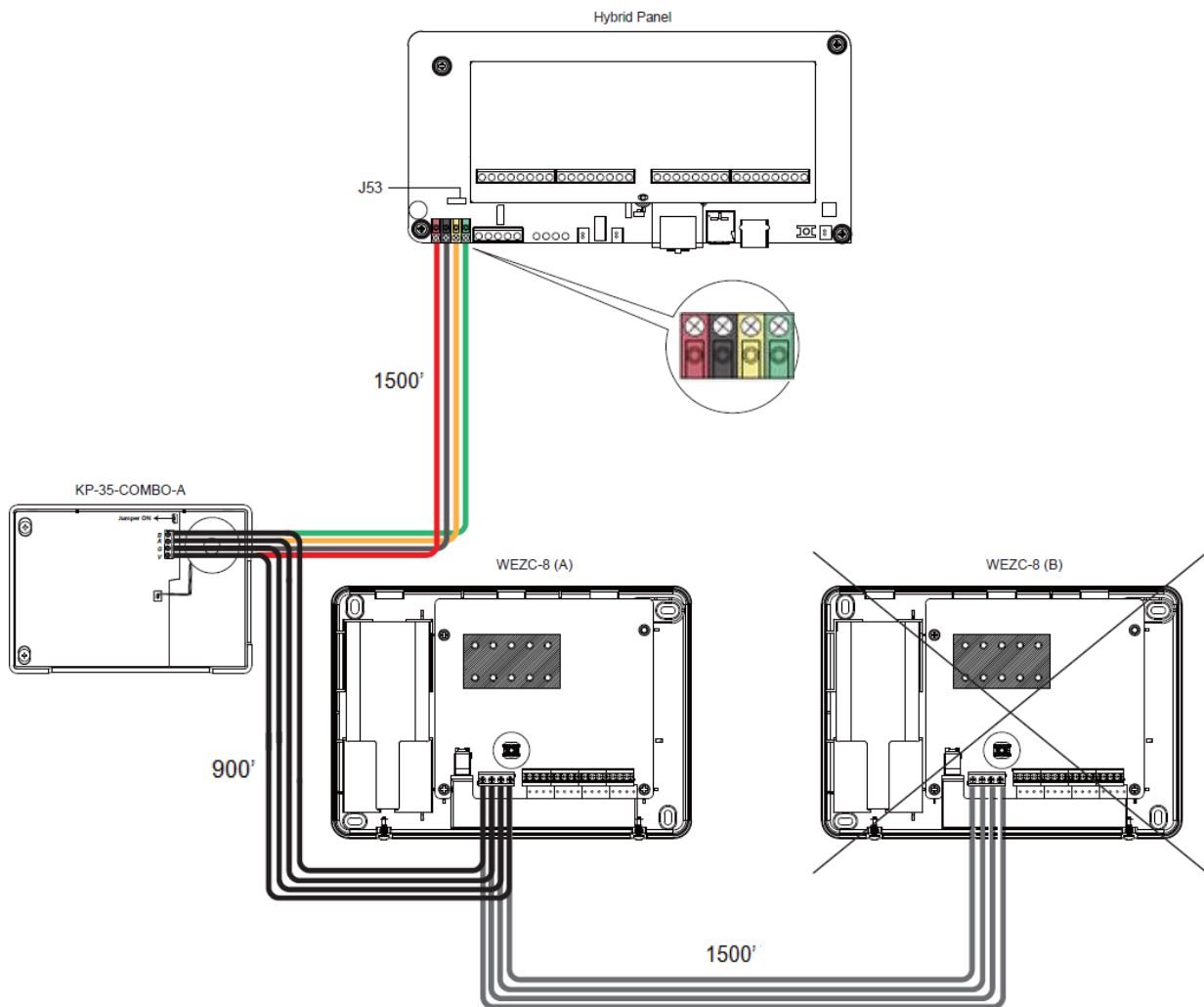
- Out of the 128 BUS devices mentioned above, a maximum of 4 keypads can be connected
- The total number of zone expansion modules, relay expansion modules, wired keypads, and other BUS devices depends on wired device's power consumption. PWB-1-BUS (power bank) or AMP-BUS (power amplifier) may be needed.

Note:

- ☞ **Please note that if you connect all the BUS devices first, then connect them all to the Control Panel to start the learning process, the maximum number of connected BUS devices should not exceed 20.**
- ☞ **If you connect more than 20 devices, the system may not operate smoothly and could cause errors on the panel programming webpage. It is recommended to connect and learn the devices one by one to ensure optimal system operation.**

Wiring Example

- KPT-35-COMBO-A is wired correctly as it is within 1500' / 456m of the panel, in wire distance.
- WEZC-8 (A) is wired correctly as it is within 2400' / 730m of the panel, in wire distance. Since it has the furthest distance from the Control Panel, ensure its jumper switch is set to ON to serve as a terminating resistor before tightening the cover fixing screw.
- WEZC-8 (B) is NOT wired correctly as it is farther than 3900' / 1185m from the panel, which makes it exceed the total length of wiring of 3000 feet.
- Ensure the furthest keypad or expansion module's Jumper Switch and Control Panel's J53 Jumper Switch are set to ON to serve as a terminating resistor.



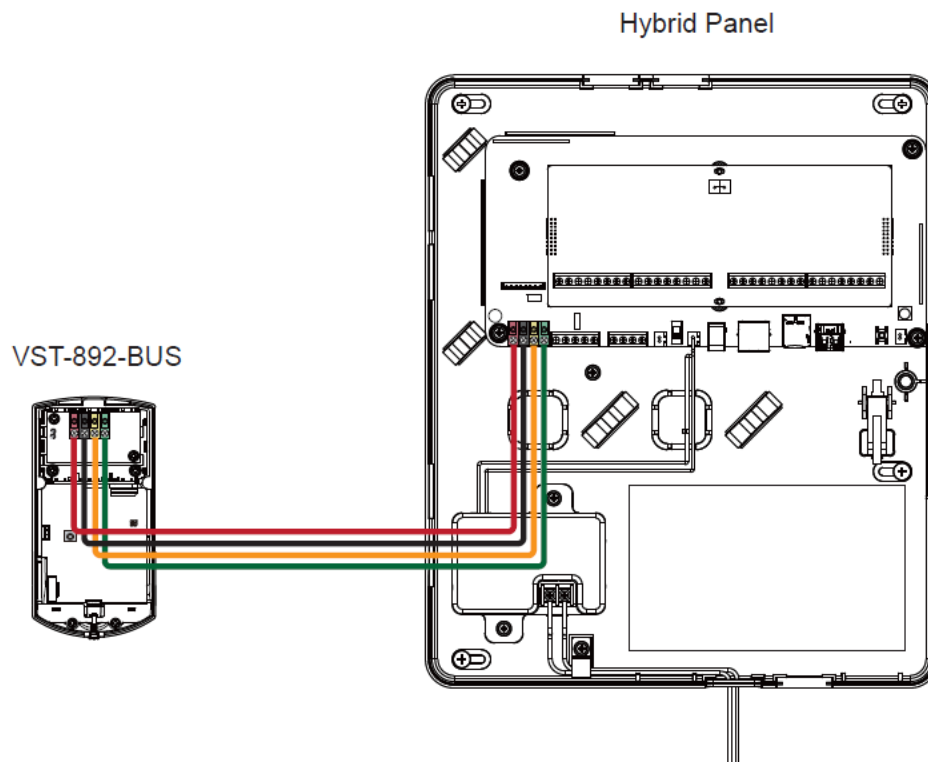
BUS Power Supply Management

- When using the Hybrid Panel to connect multiple external devices, it is important to ensure that the power supply is adequate for connected devices.
- Hybrid Panel can provide a maximum total of 13.5V/2A power supply to the connected BUS devices or expansion modules.
- Expansion boards WEZC-8B and WEPC-1B support backup battery, and they can be employed to use external power supply if power supply from the Panel is not enough.
When connecting an expansion board with external power supply to BUS, please bypass the red VDD terminal. Use the provided Wago 221 Splicing Connector to connect the VDD terminal on the Control Panel to the next BUS device that is powered by Hybrid Panel.
- Alternatively, you can use the PWB-1-BUS (Auxiliary Power Supply Module) in the data BUS system to provide power to connected BUS devices. When connecting the PWB-1-BUS to the Hybrid Panel, only connect the three terminals (GND, 485A, 485B). When connecting the PWB-1-BUS to devices that are powered by the power bank, connect all four terminals (VDD, GND, 485A, 485B).

Bus Power Supply Connection Examples

BUS devices can be connected in different combinations of devices, and be powered by different power sources. Here are three of the main possible connection methods with different device combination.

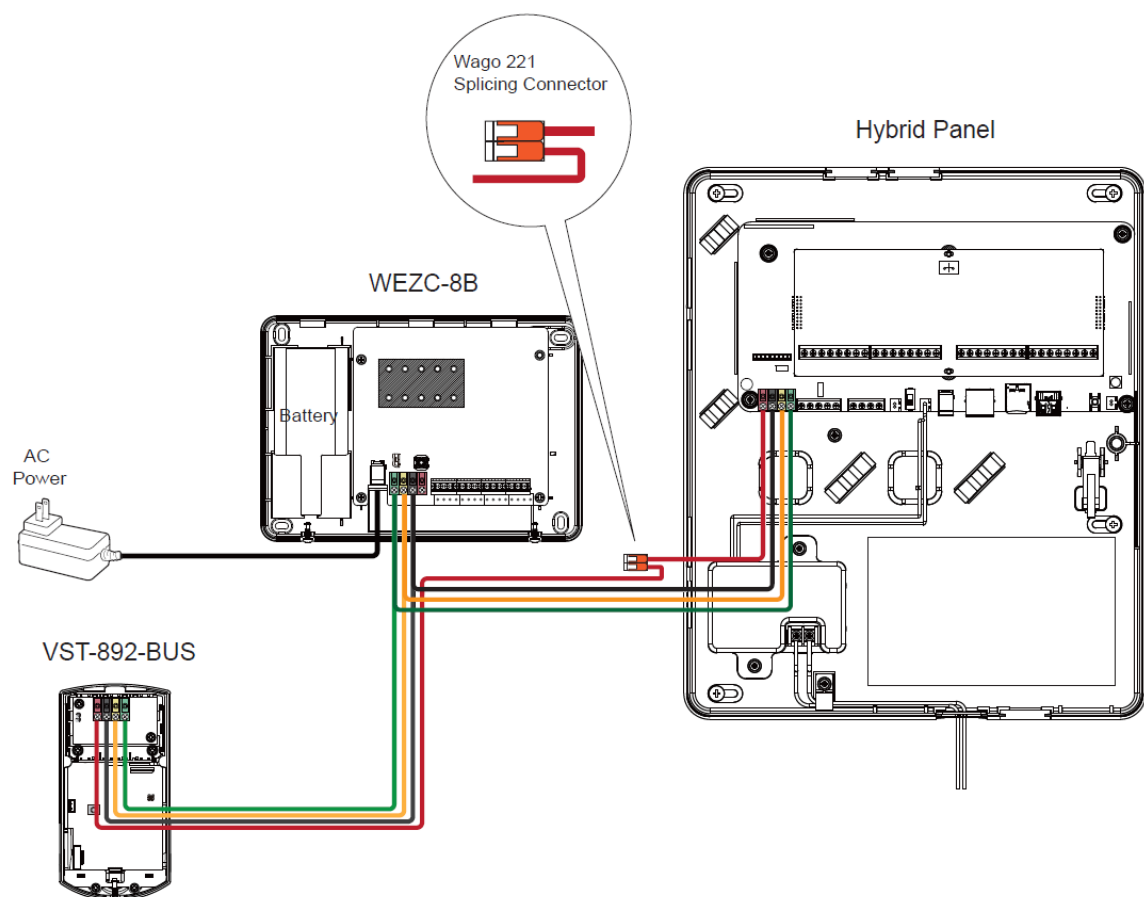
- Power supply from the Control Panel to a BUS device (VST-892-BUS):



- Power supply from the Control Panel to BUS device (VST-892-BUS), and Expansion Board (WEZC-8B) receives power from external power source:

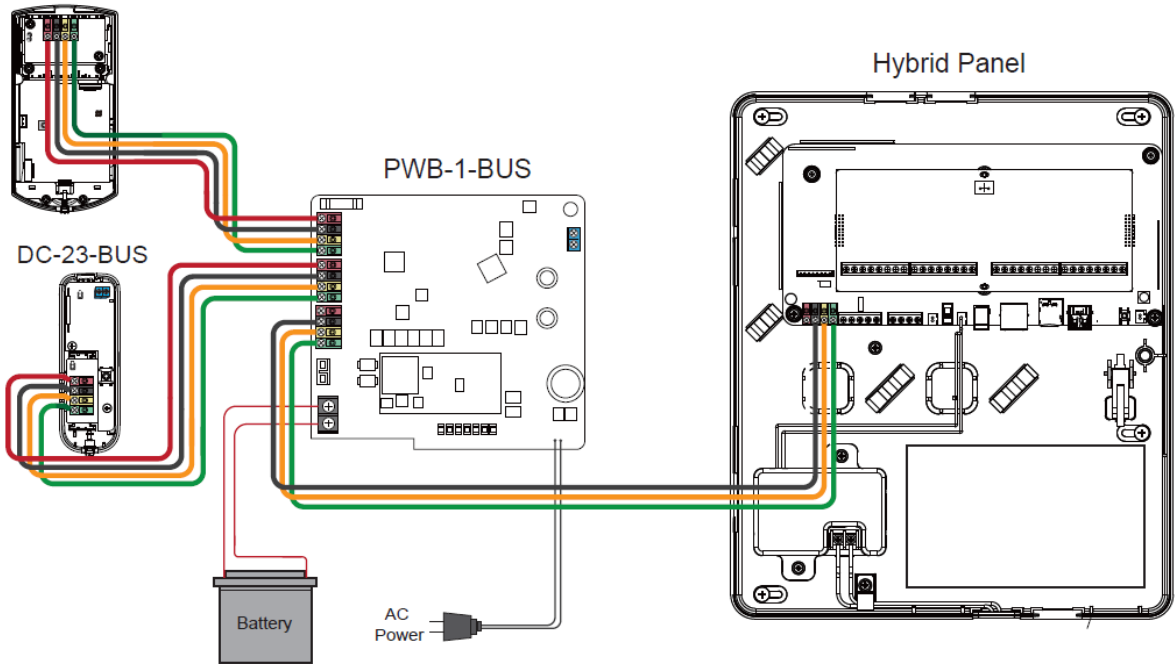
Note:

Be sure to bypass the red VDD terminal on the Control Panel using the provided Wago 221 Splicing Connector. Connect the VDD terminal to the next BUS device (VST-892-BUS as example) that is powered by Hybrid Panel.



- Power supply from PWB-1-BUS (Auxiliary Power Supply Module) to power BUS devices (VST-892-BUS, DC-23-BUS) instead of the Control Panel:

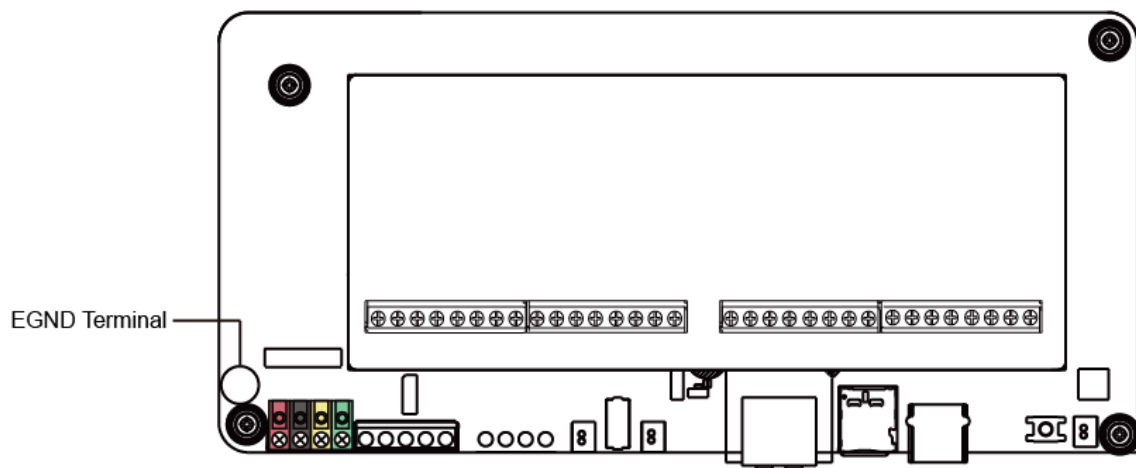
VST-892-BUS



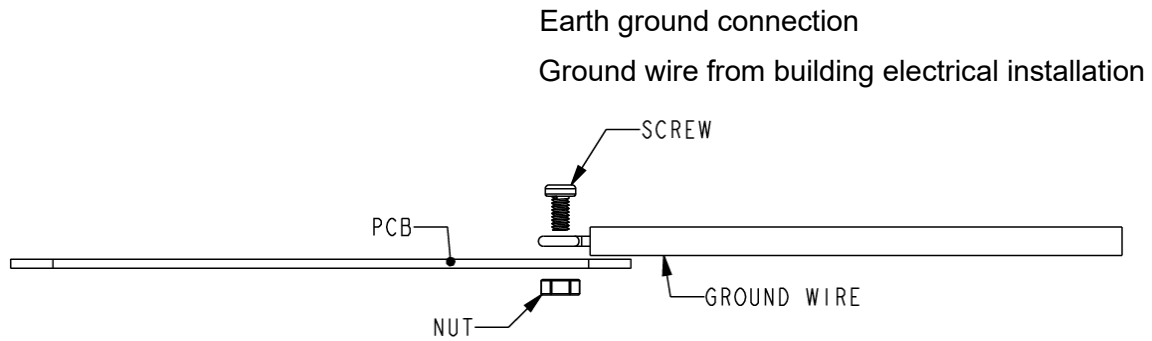
Note:

- ☞ When connecting the PWB-1-BUS to the Hybrid Panel, only connect the three terminals (GND, 485A, 485B). When connecting the PWB-1-BUS to devices that are powered by the power bank, connect all four terminals (VDD, GND, 485A, 485B).

Ground Wiring



- EGND ground wiring is implemented to protect the panel from electricity leakage.
- Prepare grounding wire (insulated green wire, minimum 22 AWG), and connect it to the building's electrical outlet. Then, secure the grounding wire onto Hybrid Panel' EGND Terminal using a ground screw and nut.

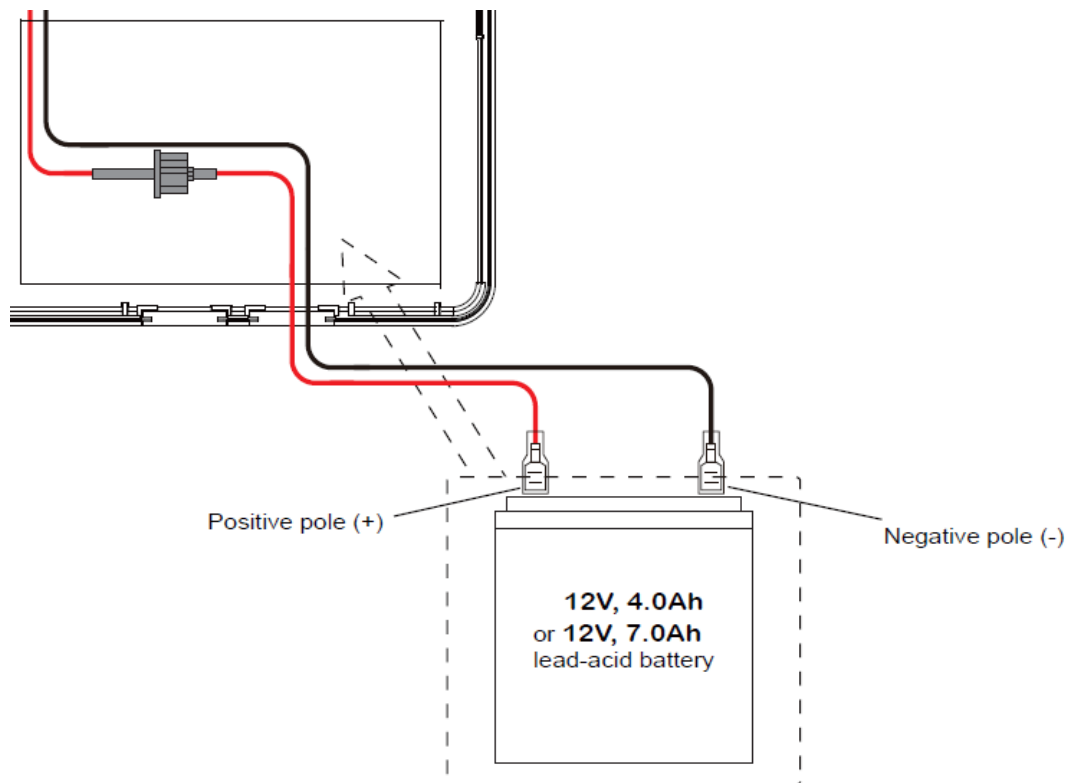


Note:

The ground wire, nut and screw are not included in the product's package.

Battery Installation

- The Control Panel can support a rechargeable battery to serve as a backup power source.
- Battery options include:
 - 1) 12V/7Ah SLA battery
 - 2) 12V/4Ah SLA battery
- To Install the battery, follow the steps below:
 - 1) Connect the GND cable (Black) to the negative pole (-) of battery.
 - 2) Connect the power cable (Red) to the positive pole (+) of battery
 - 3) Use the Velcro Tape to fix the backup battery to the Control Panel.
 - 4) Attach the battery to the hook side with Velcro Tape on the Hybrid Panel.



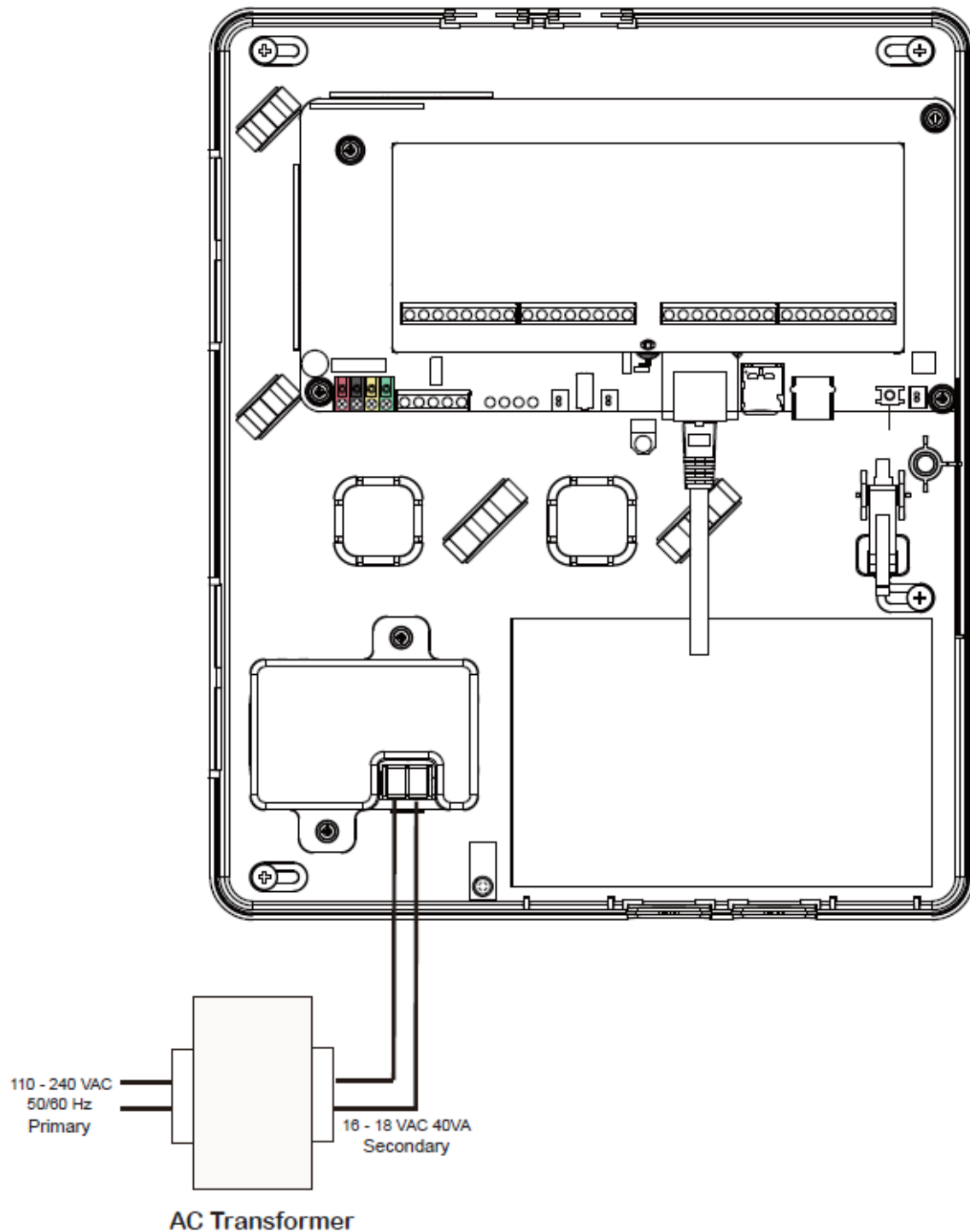
Using the built-in Power Unit

- **AC Module** built-in Power Unit or **Open Frame** built-in Power Unit is installed. For AC module, connect 2 wires of the AC Transformer to 2 terminals of the built-in Power Unit. For Open Frame, connect **BROWN** wire of the power cord to Terminal **L** of the built-in Power Unit, and connect **BLUE** wire to Terminal **N**. Please refer to two figures below.

Note:

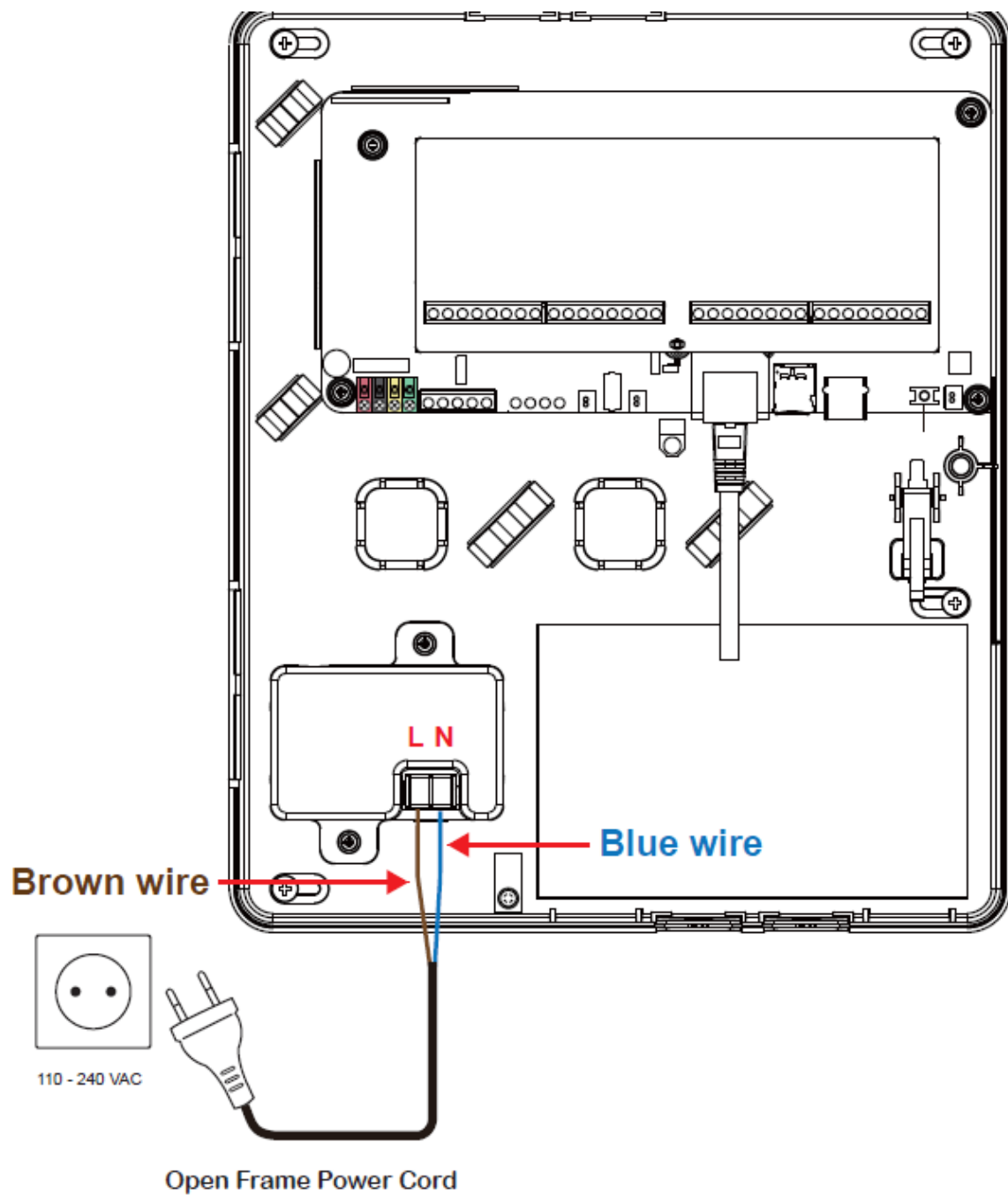
- ☞ **Ensure to turn off all power supplies including Built-in Power Unit and Battery before connecting or removing cables or wires.**

- **AC Module built-in Power Unit:**



AC Module Built-in Power Unit:
Connecting 2 wires of AC Transformer to
2 terminals of the Built-in Power Unit

- Open Frame built-in Power Unit:



Open Frame Built-in Power Unit:
Connecting **BROWN** wire to Terminal **L**
Connecting **BLUE** wire to Terminal **N**

3.4. Powering up your panel

As mentioned above, the Hybrid will either have a **16-18V AC Module** built-in Power Unit, or will have a **110-240V Open Frame** built-in Power Unit.

NOTE: To check which one is on the Hybrid you are installing, check the Built-in power unit, denoted as item number 16 on the Hybrid Diagram.

Once the main power has been wired correctly, complete the following steps:

1. Locate the customer's router.
2. Open the Hybrid using a small Phillips head screwdriver.
3. Connect the panel to the router with an Ethernet cable.
4. Connect the power adapter to a wall outlet and the other end to the panel A/C jack.
5. The LEDs on the front of the panel will illuminate and the panel will emit a two-tone beep to indicate it is on and powering up.
6. After the panel has successfully powered on, press and hold the Auto-Learn Button on the panel for 3 seconds to initiate a Manual Phone Test from the panel and initiate communications with Alarm.com. **Note: An Alarm.com customer account should be created before doing this step.**
7. Switch the battery into the *ON* position. The battery will be in the *OFF* position when shipped from the factory.
8. Close the Hybrid and tighten the screws

LED Status

1. As the panel powers on, all LEDs will illuminate then turn off to indicate the panel is powering up.
2. Once the panel is powered on, LED 3 will flash ORANGE to indicate the panel is registering with the Cellular network and acquiring an IP address from the router.
3. Once the panel successfully registers with the Cellular Network and acquires an IP address, LED 1 will blink GREEN every 5 to 10 seconds to indicate communication has been established.
 - a. See Appendix A LED Guide for a comprehensive guide to LED behavior.
4. The Cellular registration process may take up to 10 minutes. If after 10 minutes, LED 3 is still ORANGE, verify there are no fault conditions on the panel. If there are no other fault conditions, see Appendix B Communication Troubleshooting.

Button Functionality:

The reset button is located on the right side of the PCB, located inside the panel (labeled as item 24 on the diagram key). The reset button can be used for multiple occasions by holding down the button for various lengths of time:

Phone Test (3-10 seconds): Phone test is done to send a communication test from the panel

to the back end. It initiates communication and verifies that the module is communicating on the cellular network.

Learn Mode (10-20 seconds): Using the button to send the panel into Learn Mode is done to add sensor to your panel locally.

Power Cycle (20-30 seconds): A power cycle turns the security panel completely off and then powers it back on to refresh it. This is a common action taken for a variety of trouble shooting problems.

3. Adding Sensors and Peripherals

Security Sensors

Security Sensors can only be added to the system locally via a Learn Mode based sensor identification.

Adding Security Sensors using the Dealer Site or MobileTech:

1. Put the panel in learn mode
 - a. Dealer Site: *Equipment > Sensors > Add a Sensor*
 - b. MobileTech *Equipment > Sensors > Add Sensor*
2. Wait for the message “Checking for new sensors on the network.”
3. LED 1 and LED 2 will turn solid GREEN and a one-tone beep will sound.
4. Trigger a device to add it to the network.
 - a. Refer to device installation manual for how to trigger each specific device
5. When a device is added successfully, the panel will emit a one-tone beep and the device will be shown on the screen.
6. Continue triggering other devices that need to be learned in. Several devices can be added without needing to re-enter Learn Mode.
 - a. If no new devices are triggered, Learn Mode will time out after 5 minutes
7. Once all devices have been added, choose to EXIT Learn Mode.
8. On the subsequent screen, the option to define the Sensor Name, Group Number, Partition, and Sensor Activity Monitoring settings are available.
 - a. NOTE: Certain devices will require you to put the device into learn mode again in order for the device to accept new settings such as sensitivity and partition management. Please refer to device specification manual for instructions on how to complete these steps.
 - b. Sensor Groups will have a number and a short description of the behavior. For a full description of group behavior, please refer to Appendix D.

* Note the keypad must be installed within the protected premises for UL Compliant installations.

Removing Security Sensors using the Dealer Site or MobileTech

- Navigate to the Delete Sensor Page.
 - a. Dealer Site: *Equipment > Sensors > Delete a Sensor/Peripheral*
 - b. MobileTech: *Equipment > Sensors > Delete Sensor*
- Check off the DELETE box next to the sensor that needs to be removed.
- Click Send Command to remove the sensor from the system.
- The sensor will be removed from the system within 5 minutes.

Changing Sensor Settings and Configuration:

1. Dealer Site: From *Equipment* page select the appropriate sensor and action:

- a. Change Sensor Name
 - b. Change Sensor Group
 - c. Change Sensor Partition
 - d. Change Sensor Activity Monitoring Settings
2. MobileTech: Navigate to *Equipment > Sensors*.
 - a. This will allow updating the Sensor Name, Group, and Activity Monitoring Setting.
 - b. To update the Partition, continue to *Advanced Sensor Settings*.

NOTE: If your system requires compliance with the local regulation PD 6662, it is required that the system cannot be disarmed via a user code input into a keypad. Rather, the user must be provided option to disarm the system via a button press, keyfob, or prox tag that is used in tandem with an acceptable tag reading device.

Z-Wave Sensors:

Depending on the Hybrid you have, Z-wave/Automation may or may not be present on the panel.

Adding a Z-Wave device using the Dealer Site or MobileTech:

1. Put the panel into Add Mode
 - a. Dealer Site: Navigate to *Equipment > emPower tab > Advanced Z-Wave Settings > Add Z-Wave Devices*
 - b. MobileTech: Navigate to *Equipment > Z-Wave Devices > Add Z-Wave Devices*
2. Trigger Z-Wave device to add it to the network
3. When a device is added successfully, the panel will emit a one-tone beep and the device will be shown on the screen
4. Continue triggering other devices that need to be learned in. Several devices can be added without needing to re-enter Learn Mode
5. When all Z-Wave devices have been added, exit Add Mode and run a Network Rediscovery

Removing a Z-Wave Device using the Dealer Site or MobileTech

1. Put the panel into Delete Mode
 - a. Dealer Site: Navigate to *Equipment > emPower Devices tab > Advanced Z-Wave Settings > Delete Z-Wave Devices*.
 - b. MobileTech: Navigate to *Equipment > Z-Wave Devices > Delete Z-Wave Devices*.
2. Trigger the device to delete if from the network and clear the device memory.
3. Continue removing other devices that need to be removed, devices that are removed will show up on the screen.
4. Once all devices are removed, exit Delete Mode and run a network rediscovery.

Local Learning:

Devices can be learned in locally without the Alarm.com Website as well. This action is limited to security sensors and cannot be used for Z-wave devices.

1. Press and hold the Auto Learn button for 10 seconds until the panel emits a one-tone beep to indicate it has entered Learn Mode. The top LEDs will turn solid GREEN to indicate the panel is in learn mode.
2. Trigger each device following the procedure in the device installation manual.
3. When the panel receives a signal from the device it will emit a two-tone beep to indicate that it was successfully learned into the panel.
4. When all devices are learned in, press and hold the Auto Learn button on the panel for one second to exit learn mode. The panel will emit a two-tone beep and the LEDS will turn off to indicate it has returned to its normal mode.
5. Configure the sensors from the Alarm.com Dealer Site or MobileTech.

4. Hardwired Sensors: Installation and Configuration

Climax supports a vast array of hardwired sensors on the Hybrid and Hybrid Lite security panels. Hardwired sensors include:

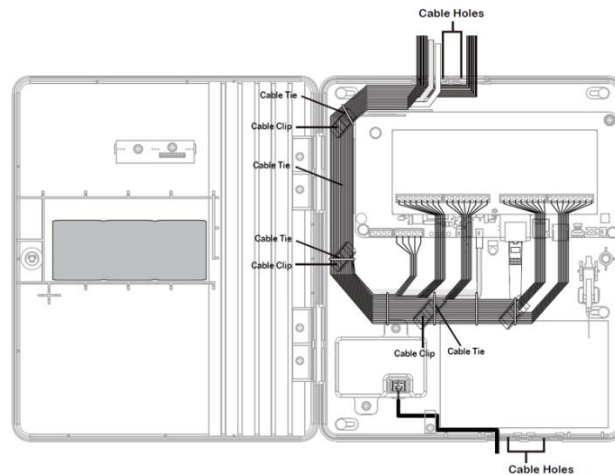
- Standard hardwired sensors that are present on a property during a takeover
 - a. These utilize the loop diagrams mentioned below for configuration
- BUS devices: These are climax branded wired devices that have additional functionality built in and use the BUS port located on the panel
 - a. This also includes expansion boards, allowing for more standard hardwired sensors to be learned into the system
 - b. The list of supported BUS devices includes:
 - i. KP-35 BUS (also the KP-35 Combo)
 - ii. WEZ 12/24/36/48 BUS
 - iii. Expansion boards
 - iv. PIR 892 BUS
 - v. IR-35 BUS
 - vi. DC-23 BUS
 - vii. SR-35 BUS
 - viii. BX-32 BUS
 - ix. PWB-BUS

Below are the high-level steps for configuring wired sensors for the Hybrid and Hybrid Lite. Please use the sections that follow to complete step 2:

Step 1. For configuration and operation of the Control Panel via Ethernet, connect an Ethernet cable to RJ-45 port.

Step 2. Complete wiring for Zone 1-16 terminals, GND terminals, AUX power terminals, PGM terminals, Bell terminals, and Keypad/BUS/Expansion board terminals. (Please see the following sections for details.)

Step 3. Arrange all wires along the wire clips and then route the wires to the two cable holes on top. Use cable ties (not included) to thread through the wire clips and secure the wires. Use the U-shaped grommets around the cable holes to manage wires.



Step 4. Connect the battery to the PCB board. See **Battery Installation** in the following section for more details.

Step 5. Connect to the mains power using the built-in power unit.

Step 6. Slide the battery switch to ON position.

5.1 Learning In standard hardwired sensors

Before progressing with the learn-in process, please ensure all standard hardwired sensors are wired in properly.

1. Log into the Partner Portal.
2. Find the customer account.
3. Click **Equipment**.
4. Click **Sensors**.
5. In *Sensor Management*, click **Add A Hardwired Sensor**.
6. Enter the hardwired sensor information:

To enroll a sensor manually, please enter the Sensor ID, Group, Sensor Name, and Code. When you are finished, click 'Add Sensor'. Commands will take effect in approximately 5 minutes. If you are attempting to add a BUS device, do not fill out the information below, and instead click the 'Add a BUS Device' to be taken to the Auto-Learn portal.

New Sensor Information

Sensor ID
(4-80)

Port Location
Hybrid

Port Number
1

Loop Number
1

Resistor
No EOL

Group
91 - 24 Hour Silent Water

Sensor Type
Not Selected

Partition
1

Sensor Name

Add Sensor

Scan For Expansion Modules

Add a BUS Device

Sensor Group Definition- [click here for HxGW sensor groups.](#)
Hardwired Sensor Guide - [Click here for the Hardwired Installation Guide](#)

- Sensor ID: Dictates what zone the sensor learns into
- Port location: Refers to the where the hardwired sensor is wired in. If your install

deals with Expansion boards, ensure the Expansion board is learned in properly by going through the BUS Device learn-in process before wiring up standard hardwired sensors.

- Once an expansion board has been learned in, it will populate under this section
 - Port Number: Refers to the hard wired port on the Hybrid or expansion board you are wiring the hardwired sensor into
 - If you are working with expansion boards, this section will update to reflect available ports on the expansion board once it is learned in and is selected in the "port location" section
 - Loop Number: Refers to the wiring configuration of the sensor. Please refer to the loop guide listed above to determine which number loop configuration should be used.
 - **NOTE: Loops 1 and 2 do not have EOL resistors in them**
 - Resistor: Refers to the value of the resistor that you are using in your loop
 - Note: For loops using multiple resistors, the resistor values must be identical.
 - Group: This defines how a sensor operates within the system and how it responds when triggered. Use the link below the configurable boxes to see a list of available sensor groups for the sensor you are adding.
 - Sensor type: Needed to determine what type of sensor is being learned in. The options in the drop down will be:
 - Door Contact
 - Carbon Monoxide
 - Smoke Detector
 - IR
 - Water Sensor
 - Glass Sensor
 - Heat Detector
 - Partition: Determines which partition the sensor is learned into
 - Sensor name: Will determine what the sensor name will be that is stored on the panel
1. Once all information is entered, click **Add Sensor**.
 2. Verify the sensor appears on the *Equipment* list page, and perform a walk test.

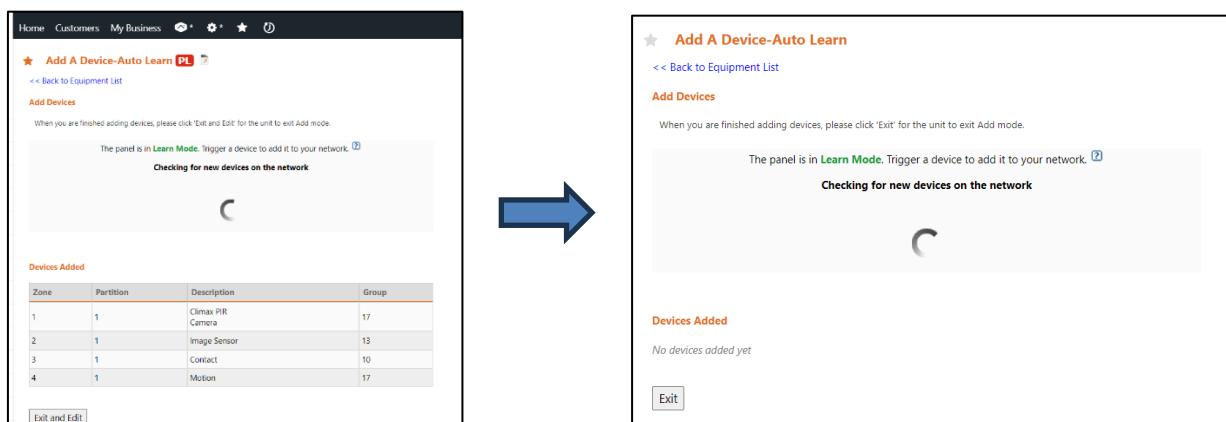
5.2 Editing Standard Hardwired Sensors

- Currently, editing a standard hardwired sensor is not supported. Please follow the steps below to change the setting of a hardwired sensor:
 - Delete the sensor from the account

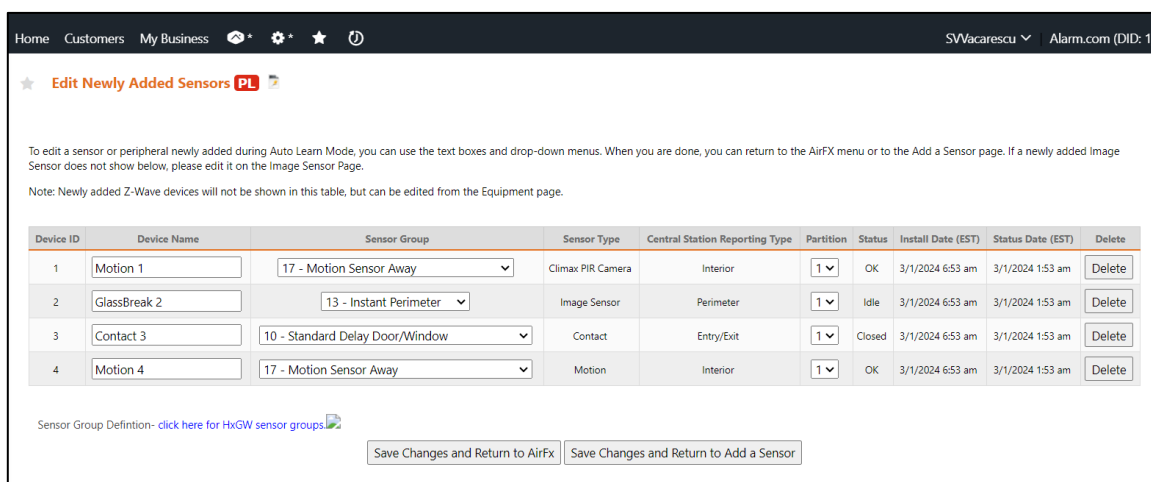
- Re-learn the sensor with the new information that is needed

5.3 Learning in BUS Devices

- To learn in a BUS device, please first ensure that it is wired into the Hybrid or into another BUS device properly.
- Go to the auto-learn feature on the dealer site or mobile tech and put the panel into learn mode. If they are wired in correctly, BUS Devices should auto-populate on the page
 - If they don't auto-populate, please continue onward to check the equipment page to see if they populated there.



- Once they populate, you can edit the BUS devices as if they were regular sensors



5.3 Editing BUS Devices

Editing a BUS devices is done similar to the way wireless sensors are edited/deleted

5.4 PGM Support

- PGM Support is currently not supported by Alarm.com

5.5 Identifying a BUS Device

- On the Sensor list on the Equipment Page, you can expand all columns to reveal a a “Wiring Configuration” column, indicating if the device is a BUS device

Sensor List:

[Hide Extra columns](#)

Device ID 	Device Name	Sensor Group	Sensor Type	Central Station Reporting Type	Partition	State	Sensor Activity Monitoring	Sensor Activity Monitoring Eligible?	Serial Number	Wiring Configuration	Dr
1	Contact 1	10	Contact	Entry/Exit	1	Bypassed	Off	Yes	01782E1400	Wireless	1
2	Motion 2	17	Climax PIR Camera	Interior	1	Tampering	Off	Yes	IN:030c2800	BUS	1/
3	DC wired Hybrid 1	10	Contact	Entry/Exit	1	OK	Off	Yes	IN:01	Hardwired	1

- On the equipment page, hit "change" to open up a the sensor specific command menu. In the dropdown, select, "identify" and send the command. This will cause the BUS device to activate an identifier such as a buzzer or LED

5.6 Enrolling hardwired BUS Keypads

- Power off the Hybrid/Hybrid Lite panel.
- Wire in the KP-35.
- Power on the panel.
- Go to *Remote Toolkit* and search for *Installer Code*.
- Enter **[7982]**.
- Displayed on the keypad should be the following prompt: *Choose Area & Zone And Enter Codes*.
- Select the *Area* = partition.
- Select the *Zone* = sensor zone.
- Displayed next on the keypad should be: *Enter Pin Code*.
- Enter the master pin code of the panel.
- Displayed next on the keypad should be: *Enter Installer Code*.

Important: Do not put the installer code in without completing the next step.

12. Put the panel into learn in mode.
13. Enter the installer code **[7982]**.
14. The keypad should display *Added Keypad to Area # Zone #*, where the area and zone are the partition and sensor zone selected in the preceding steps.
15. The keypad should populate on the auto-learn page.

5. Walk Test and Device Mounting

Once all devices are learned in, place the devices in their final locations and run a walk test to ensure that all devices are signaling properly in their final locations. Hardwired sensors may already be placed depending on the type of install occurring.

Walk Test

1. Enter Walk Test mode, LED 1 and LED 2 will blink GREEN while in Walk Test mode
 - a. Dealer Site: *AirFx Remote Toolkit > Walk Test*
 - b. MobileTech: *Equipment > Sensors > Walk Test*
2. Select Begin Walk Test and activate each sensor
3. As sensors are activated, their action will appear onscreen and the sensor will emit a two-tone beep
4. After all sensors have been triggered, select End Walk Test to exit walk test mode.
 - a. If a sensor trigger did not register on the Walk Test, try moving it closer to the control panel and trigger it again. If the device works then, try using an RF repeater to extend the range of the control panel.

Device Mounting

Once devices are signaling in their final locations, they can be mounted according to the instructions provided with each device.

After finish all setting, click **OK** to confirm change.

6. System Programming

The system settings can be programmed via the *Remote Toolkit* on the Alarm.com Dealer Website or MobileTech App. The below outlines the definitions of the various system settings along with descriptions and setting options including the system default.

- **AC Fail Report:** When an AC power failure is detected, your Control Panel will report to the Central Monitoring Station according to the duration set under AC Fail Report. If 5 minutes is set, the event will be automatically reported to the CMS after 5 minutes. Your Control Panel will start to use its battery power instead of the mains power until the fault even is cleared.
- **Jamming Report:** Jamming period is specified as background RSSI level detected exceeding the threshold for a period of time. The jamming period detected will be accumulated.
- 3 options: Disable, 1 minute and 2 minutes are provided. If 1 minute is set, once the total jamming period exceeds 30 seconds within 1 minute, a “Jamming” message will be reported to the Central Monitoring Station. If 2 minutes is set, once the total jamming period exceeds 60 seconds within 2 minutes, a Jamming message will be reported to the Central Monitoring Station. If Disable is set, the control panel will not send a jamming report to the Central Monitoring Station if a jamming fault is detected.
- **Auto Check-in Interval:** this is to select whether the Control Panel needs to send check-in reporting to the Central Station automatically and to select the period of time between check-in reports. Options available are **Disable**, **1 hour**, **2 hours**, **3 hours... up to 4 Weeks**.
- **Auto Check-in Daily Time:** When the Auto-check in Interval is set to 24HR, this setting allows the user to specify the exact time of day when the automatic check-in process should occur. For example, if the user sets the daily time to 3:00 PM, the security panel will automatically send a check-in signal to the central monitoring system every day at that specific time.
- **IR Camera Resolution of Alarm Images:** This is to select the resolution and number of pictures taken by PIR Camera when the camera detects a movement in armed mode.
- **Outdoor IR Camera in Grayscale:** This is to select whether pictures from Outdoor PIR Camera should be taken in grayscale instead of color pictures.
- Options available are: **Disable** (Color Picture) and **Enable** (Greyscale picture)
- **Bypass Ethernet:** Select to enable or disable Bypass Ethernet function. When **YES** is selected, the Control Panel will bypass connection fault when Ethernet cable unplugged status is detected.
- **Service Failure Reporting:** This setting allows to periodically check the connectivity to a specific server, such as a Google server, in order to assess the status of the internet connection and report any service failures.
- **Power supply overcurrent restart time:** If a short circuit is detected at the output BUS or wired zone terminals, the Hybrid Panel will stop supplying power to all voltage output

terminals, including the 8 wired zone auxiliary output, PGM, and BUS VDD terminal, to ensure that the panel CPU is protected and will continue to operate without interruption.

- You can set the power supply restart time to 1, 3, 5, 10, or 15 minutes. The Panel will restart supplying power according to the duration set.
 - If a short circuit status is still not cleared when re-supplying power, the Panel will cut power for the voltage output terminals again.
 - Hybrid Panel will report to the Central Monitoring Station when a short circuit is detected. When power output is restored, the panel will also make report to the Central Monitoring Station.
- **Mute Internal Siren:** The setting allows users to disable the internal siren of the security system and instead emit a beep sound for audible notifications.
- **Final Door:** If set to **On**: When the system is Away Armed and under exit timer countdown, if a opened Door Contact set to Entry attribute is closed, the system will automatically arm the system even if the exit delay timer has not expired yet.
- **Arm Fault Type:** Select how the system should respond when it is being armed under fault condition.
 - Confirm: The panel will first display a “Mode Change Fault” message and emit 2 beeps. Arming again within 10 seconds will force arm the system.
 - Direct Confirm: The system will be force armed directly without displaying fault message and report an event.
- **Tamper Alarm:** Select whether the siren should sound alarm when the tamper is triggered.
 - Full Arm: when tamper is triggered under Full arm mode, Control Panel raises a local alarm and sends report to the monitoring center. While under Home Arm or Disarm modes no alarm will be activated, nor report sent.
 - Always: Control Panel raises a local alarm and send report for tamper-trigger in all modes.
- **Supervision Check:** Select to enable or disable system supervision function. When **ON** is selected, the Control Panel will monitor the accessory devices according to the supervision signal received. **Time Setting**
- **Supervision Timer:** The Control Panel monitors accessory devices according to the supervision signal transmitted regularly from the device. User this option to set a time period for receiving supervision signals. If the Control Panel fails to receive supervision signal from a device within this duration, it will consider the device out of order and report the event accordingly.
- **Entry Delay 1 for Full Arm:** Set Entry Delay Timer 1 for full arm mode. When a sensor set to Start Entry Delay 1 is triggered under Full Arm mode, the control panel will begin Entry Delay Timer countdown according to duration set with this option
- If the Control Panel is disarmed before the Entry Delay Timer expires, the panel returns to Disarm mode and no alarm is activated. If the Control Panel is not disarmed before the Entry Delay Timer expires, the alarm will be activated and the panel will send report.

- **Entry Delay 2 for Full Arm:** Set Entry Delay Timer 2 for full arm mode. When a sensor set to Start Entry Delay 2 is triggered under Full Arm mode, the control panel will begin Entry Delay Timer countdown according to duration set with this option
- If the Control Panel is disarmed before the Entry Delay Timer expires, the panel returns to Disarm mode and no alarm is activated. If the Control Panel is not disarmed before the Entry Delay Timer expires, the alarm will be activated and the panel will send report.
- **Exit Delay for Full Arm:** Set the Exit Delay Timer when entering Full Arm mode. When the user changes system mode to Full Arm, the panel will begin Exit Delay Timer Countdown and enter Full Arm mode when the timer expires. The user must leave area protected by sensors before the timer expires, otherwise an alarm will be activated with the sensor is triggered.
- **Entry Delay 1 for Home Arm:** Set Entry Delay Timer 1 for Home Arm mode. When a sensor set to Start Entry Delay 1 is triggered under Home Arm mode, the control panel will begin Entry Delay Timer countdown according to duration set with this option
- If the Control Panel is disarmed before the Entry Delay Timer expires, the panel returns to Disarm mode and no alarm is activated. If the Control Panel is not disarmed before the Entry Delay Timer expires, the alarm will be activated and the panel will send report.
- **Entry Delay 2 for Home Arm:** Set Entry Delay Timer 2 for Home Arm mode. When a sensor set to Start Entry Delay 2 is triggered under Home Arm mode, the control panel will begin Entry Delay Timer countdown according to duration set with this option
- If the Control Panel is disarmed before the Entry Delay Timer expires, the panel returns to Disarm mode and no alarm is activated. If the Control Panel is not disarmed before the Entry Delay Timer expires, the alarm will be activated and the panel will send report.
- **Exit Delay for Home Arm:** Set Exit Delay Timer for Home Arm mode. When the user changes system mode to Home Arm, the panel will begin Exit Delay Timer Countdown and enter Home Arm mode when the timer expires. The user must leave area protected by sensors before the timer expires, otherwise an alarm will be activated with the sensor is triggered (Default as 10 seconds).
- **Alarm Length:** Set the duration the external siren should sound when an alarm is activated.
- **Cross Zone Timer:** Please refer to *10.3 Cross Zone Timer* for details
- **Fire Verification Time:** Please refer to *10.4 Fire Verification Timer* below for details.
- **Sound Setting**
- **Door Chime Setting:** this function is available only when the attribute of Door Contact (DC) and/or PIR detector (IR) is set as **Door Chime**.
- The Control Panel sounds a Door Chime (Ding-Dong Sound) while the DC and/or IR is activated in Disarm / Full / Home / Entry mode.
- **Entry Delay Sound for Full Arm:** this is for you to decide whether the Control Panel sounds count-down beeps and volume of beep during the entry delay time in the full arm mode.
- **Exit Delay Sound for Full Arm:** this is for you to decide whether the Control Panel

sounds count-down beeps and volume of beep during the exit delay timer in the full arm mode.

- **Entry Delay Sound for Home Arm:** this is for you to decide whether the Control Panel sounds count-down beeps and volume of beep during the entry delay time in the home arm mode.
- **Exit Delay Sound for Home Arm:** this is for you to decide whether the Control Panel sounds count-down beeps and volume of beep during the exit delay timer in the home arm mode.
- **Confirm Sound:** this is for you to decide whether to turn off/or adjust Control Panel beeping sounds when changing Arm/Home Arm/Disarm mode.
- **Warning beep:** this is for you to decide whether the Control Panel will sound a warning beep whenever a fault condition has been detected and displayed. The warning beep will be silenced after the Fault message has been read by the user. When a new fault condition is detected, it will then again emit a warning beep every 30 sec.
- **Entry/ Exit Only Final Beeps:** This is for you to determine when the Control Panel should start warning beep during Entry or Exit countdown timer. For example, if the setting is set to 5 seconds, the Control Panel will only stat warning beep during the last 5 seconds of Entry or Exit countdown timer. When set to Disable, the Control Panel will sound warning beep during the entire Entry or Exit countdown timer.

7. Remote Toolkit

Remote toolkit is where numerous trouble shooting and configuration tools are found. From the Remote Toolkit, you can:

- Run a System Check to ensure everything is running properly
- Turn on/off trouble beeps
- Receive a list of active trouble conditions affecting the panel
- A summary of the communication performance
- A summary of sensor connectivity
- A snapshot of customer/dealer engagement

The “Toolkit actions” is also found here. Under the Toolkit Actions tab, you can:

- Add a device
- Schedule an appointment
- Log in as customer
- Pause notifications
- Acknowledge trouble beeps
- View panel settings
- View trouble conditions/trouble condition history

Lastly, the Command Catalog will have everything you need to trouble shoot a customer’s system.

8. Appendix

9.1. Appendix A: LED Guide

LED 1 (Red/Green)

Red ON – Partition One is Armed Away

RED Flash – Partition One is Armed Stay

Green On – System is in Auto Learn Mode

Rapid Green Flash – System is in Walk Test/Z-Wave Delete Mode

Green Flash every 5 Seconds – Panel is Communicating over both Cell and IP Paths

Green Flash every 10 seconds – Panel is communicating over 1 path

LED 2 (Red/Green)

Red ON – Partition Two is Armed Away

RED Flash – Partition Two is Armed Stay

Green On – System is in Auto Learn Mode

Rapid Green Flash – System is in Walk Test/Z-Wave Delete Mode

LED 3 (Red/Yellow)

Red On – Alarm in memory

Red Flash – Alarm

Orange On – System fault.

Orange Flash – Registering to Cellular Network

Possible System Faults:

- Sensor Tamper
- Sensor Malfunction
- Sensor Low Battery
- Panel Tamper
- Panel Low Battery
- Panel Battery Off
- Cell Communication Failure
- Broadband Communication Failure/No Ethernet Cable Plugged In

9.2. Appendix B: Communication Troubleshooting

Cell Failure

There are several possible reasons the cellular communication is not connected to Alarm.com:

- Wireless signal strength issue or coverage issue.
 - Make sure the module is installed high enough within the structure and near an exterior wall. Alarm.com recommends that the module be installed above ground

(i.e. NOT in a basement) and not in the middle of a large structure (i.e. NOT in an interior closet far from the exterior walls).

- Make sure nothing near the module could be causing RF interference. Large metal objects or ductwork, sheet metal walls and appliances that generate RF may interfere with the module.
- If the customer is located in a borderline wireless coverage area, a high-gain, remote antenna may be required. Contact your Alarm.com Rep about this if needed.
- Make sure the SIM card has not been removed.

Broadband Failure

For issues with broadband communication, please check the following:

- Ensure the Ethernet cable is plugged into both the customer's router and the panel.
 - If the panel is connected via WiFi, ensure that the SSID and Password are correct.
- Ensure that the customer's router has power and internet by checking the internet connection of a different device on the network.
- If the panel is connected via Static IP address, ensure that all the parameters are set in accordance with network specifications.

Accessing Panel Troubleshooting Page

Via Customer Router:

1. Connect a laptop, phone, or tablet to the same network as the panel.
2. Login to customer router using the router manufacturer instructions and find the IP address of the panel.
3. Navigate to the panel's IP address

9.3. Appendix C: Panel Programming Options

Setting	Setting Description	Available Options
AC Fail Report Timer	If AC power fails, this will specify communication delay.	Disabled
		20 Minutes
		3 Minutes
		30 Minutes
		5 Minutes
		1 Hour
		1 Minute
		10 Minutes
		15 Minutes
Jamming Report Timer	Jamming Report Delay Timer	Disabled

		2 Minutes
		1 Minute
Arm Fault Type (Partition 1)	When system is being armed under fault condition. 1. Confirm: The panel will display a Mode Change Fault message. Arming again within 10 seconds will force arm the system. 2. Direct Arm: The system will be force armed directly and report an event.	Confirm
		Direct Arm
Final Door (Partition 1)	Sets whether the system will automatically Arm Away once a door contact, set as entry device, is detected as closed, even if the count-down period is not yet complete.	Disable
		Enable
Entry Delay 1 for Arm Away (Partition 1)	Entry Delay 1 – For UL Compliant installations, Entry Delay must be set to 45 seconds or less.	0 – 240 Default: 30
Entry Delay 1 for Arm Away (Partition 2)	Entry Delay 1 – For UL Compliant installations, Entry Delay must be set to 45 seconds or less.	0 – 240 Default: 30
Entry Delay 2 for Arm Away (Partition 1)	Entry Delay 2 – For UL Compliant installations, Entry Delay must be set to 45 seconds or less.	0 – 240 Default: 60
Entry Delay 2 for Arm Away (Partition 2)	Entry Delay 2 – For UL Compliant installations, Entry Delay must be set to 45 seconds or less.	0 – 240 Default: 60
Exit Delay for Arm Away (Partition 1)	Exit Delay - For UL Compliant Installations, Exit Delay must be set to 120 seconds or less.	0 – 240 Default: 30
Exit Delay for Arm Away (Partition 2)	Exit Delay - For UL Compliant Installations, Exit Delay must be set to 120 seconds or less.	0 – 240 Default: 30
Door Chime (Partition 1)	Select whether a sensor will cause the panel to chime when it is activated.	High
		Off
		Low
Door Chime (Partition 2)	Select whether a sensor will cause the panel to chime when it is activated.	High
		Off
		Low
Warning Beep (Partition 1)	Determines whether the panel will	High

	issue a warning beep for fault conditions.	Off
		Low
Warning Beep (Partition 2)	Determines whether the panel will issue a warning beep for fault conditions.	High
		Off
		Low
Entry Delay 1 for Arm Stay (Partition 1)	Entry Delay 1 – For UL Compliant installations, Entry Delay must be set to 45 seconds or less.	0 – 240 Default: 30
Entry Delay 1 for Arm Stay (Partition 2)	Entry Delay 1 – For UL Compliant installations, Entry Delay must be set to 45 seconds or less.	0 – 240 Default: 30
Entry Delay 2 for Arm Stay (Partition 1)	Entry Delay 2 – For UL Compliant installations, Entry Delay must be set to 45 seconds or less.	0 – 240 Default: 60
Entry Delay 2 for Arm Stay (Partition 2)	Entry Delay 2 – For UL Compliant installations, Entry Delay must be set to 45 seconds or less.	0 – 240 Default: 60
Exit Delay for Arm Stay (Partition 1)	Exit Delay – For UL Compliant Installations, Exit Delay must be set to 120 seconds or less.	0 – 240 Default: 30
Exit Delay for Arm Stay (Partition 2)	Exit Delay – For UL Compliant Installations, Exit Delay must be set to 120 seconds or less.	0 – 240 Default: 30
Dealer Code	Set Dealer Code to override Sensor Tamper during INCERT monitoring. Note: This setting is only used when INCERT monitoring is enabled.	Any 4 Digit User Code Default: NULL
Auto Check-in Interval	Select the desired Auto Check-In frequency from the drop-down menu below. After you have updated the setting and clicked "change", please click "Upload From Panel" at the top of the page to make sure the new setting takes effect.	Disable
		Every Week
		Every Day
		Every 6 Hours
		Every 4 Weeks
Auto Check-in Offset Period	Auto Check-in Offset	2hr
		3hr
		4hr
		6hr
		8hr
		1hr
		12hr

Dialer Delay	Dialer Delay	0 seconds
		15 seconds
		30 seconds
		45 seconds
Tamper Alarm (Partition 1)	Tamper Alarm	Arm Away
		Always
Tamper Alarm (Partition 2)	Tamper Alarm	Arm Away
		Always
Supervision Check (Partition 1)	Determines whether the panel supervises sensors on the system to ensure they are still signaling.	Off
		On
Supervision Check (Partition 2)	Determines whether the panel supervises sensors on the system to ensure they are still signaling.	Off
		On
Supervision Timer (Partition 1)	Select the frequency of the sensor supervision check.	Disable
		4 hours
		5 hours
		6 hours
		8 hours
		12 hours
		2 hours
		24 hours
Supervision Timer (Partition 2)	Select the frequency of the sensor supervision check.	Disable
		4 hours
		5 hours
		6 hours
		8 hours
		12 hours
		2 hours
		24 hours
Alarm Length (Partition 1)	Alarm Length – For UL compliant installations, this must be set to at least 4 minutes.	Disable (Not Allowed for UL Compliant Installations)
		2 minutes
		3 minutes
		4 minutes
		5 minutes
		6 minutes
		7 minutes
		8 minutes

		9 minutes
		1 minute
		10 minutes
		11 minutes
		12 minutes
		13 minutes
		14 minutes
		15 minutes
Alarm Length (Partition 2)	Alarm Length – For UL compliant installations, this must be set to at least 4 minutes.	Disable (Not Allowed for UL Compliant Installations)
		2 minutes
		3 minutes
		4 minutes
		5 minutes
		6 minutes
		7 minutes
		8 minutes
		9 minutes
		1 minute
		10 minutes
		11 minutes
		12 minutes
		13 minutes
		14 minutes
		15 minutes
Cross Zone Timer (Partition 1)	Select the time window within which 2 cross zones must be tripped to generate a verified alarm.	Disable
		2 minutes
		3 minutes
		4 minutes
		5 minutes
		6 minutes
		7 minutes
		8 minutes
		9 minutes
		1 minute
		10 minutes
		11 minutes
		12 minutes
		13 minutes

		14 minutes
		15 minutes
Cross Zone Timer (Partition 2)	Select the time window within which 2 cross zones must be tripped to generate a verified alarm.	Disable
		2 minutes
		3 minutes
		4 minutes
		5 minutes
		6 minutes
		7 minutes
		8 minutes
		9 minutes
		1 minute
		10 minutes
		11 minutes
		12 minutes
		13 minutes
		14 minutes
		15 minutes
Fire Verification Timer (Partition 1)	Select the time window within which verified fire zones must be tripped to generate a verified alarm.	Disable
		2 minutes
		3 minutes
		4 minutes
		5 minutes
		6 minutes
		7 minutes
		8 minutes
		9 minutes
		1 minute
		10 minutes
		11 minutes
		12 minutes
		13 minutes
		14 minutes
		15 minutes
Fire Verification Timer (Partition 2)	Select the time window within which verified fire zones must be tripped to generate a verified alarm.	Disable
		2 minutes
		3 minutes
		4 minutes
		5 minutes
		6 minutes

		7 minutes
		8 minutes
		9 minutes
		1 minute
		10 minutes
		11 minutes
		12 minutes
		13 minutes
		14 minutes
		15 minutes
Entry Delay Sound for Arm Away (Partition 1)	Entry Delay Sound Volume for Arm Away	High
		Off
		Low
Entry Delay Sound for Arm Away (Partition 2)	Entry Delay Sound Volume for Arm Away	High
		Off
		Low
Exit Delay Sound for Arm Away (Partition 1)	Exit Delay Sound Volume for Arm Away	High
		Off
		Low
Exit Delay Sound for Arm Away (Partition 2)	Exit Delay Sound Volume for Arm Away	High
		Off
		Low
Entry Delay Sound for Arm Stay (Partition 1)	Entry Delay Sound Volume for Arm Stay	High
		Off
		Low
Entry Delay Sound for Arm Stay (Partition 2)	Entry Delay Sound Volume for Arm Stay	High
		Off
		Low
Exit Delay Sound for Arm Stay (Partition 1)	Exit Delay Sound Volume for Arm Stay	High
		Off
		Low
Exit Delay Sound for Arm Stay (Partition 2)	Exit Delay Sound Volume for Arm Stay	High
		Off
		Low
Entry/Exit Only Final Beeps (Partition 1)	When selected, the Entry/Exit beeps will only sound for the last 3-10 seconds of Entry/Exit Delay.	Disable
		10 seconds
		3 seconds
		4 seconds
		5 seconds
		6 seconds

		7 seconds
		8 seconds
		9 seconds
Entry/Exit Only Final Beeps (Partition 2)	When selected, the Entry/Exit beeps will only sound for the last 3-10 seconds of Entry/Exit Delay.	Disable
		10 seconds
		3 seconds
		4 seconds
		5 seconds
		6 seconds
		7 seconds
		8 seconds
		9 seconds
Swinger Shutdown	Swinger Shutdown. When set to 0, setting is off.	0 – 255 Default: 0
Ethernet Bypass	Ethernet Bypass. When enabled, the Ethernet cable being unplugged will not cause a fault at the panel.	Disable
		Enable
Access Code Digits	Set user access codes to be 4 or 6 digits.	4 Digits
		6 Digits
PIR Camera Image Resolution	Choose what resolution your PIR Camera images are uploaded at as well as how many images are uploaded automatically during alarm.	3xqVGA
		1xqVGA
		1xVGA
External Siren Tamper Sound	Choose whether an external siren learned into the control panel sounds when it is physically tampered. Recommended if performing maintenance on the device. Note: This setting is download only.	Off
		On
External Siren Confirmation Sound	Choose whether an external siren sounds a confirmation beep to indicate a change in arming state. Note: This setting is download only and may change due to Silent Arming selection.	Off
		On
External Siren Entry/Exit Sound	Choose whether an external siren sounds during entry or exit delay. Note: This setting is download only and may change due to Silent Arming selection.	Off
		On
RF Replay	Enable/Disable RF Replay attacks	Enable/Disable

	protection	
Duress Code	Enable/Disable duress code	Enable/Disable
Mute Internal Panel Siren	This allows the panel internal siren to be turned off. It is recommended to have an external siren associated with the system before doing this. Enabled = siren is muted. Disabled = Siren is not muted. Default value: Disabled.	Enable/Disable
Ignore Sensor Tamper	Enabling this will have the panel ignore sensor tamper that are registered from the field. This is a panel-wide setting and will ignore sensor tamper on both Partition 1 and Partition 2 if enabled.	Enable/Disable
External Antenna Tamper	Determines whether the tamper from an external antenna is enabled or disabled.	Enabled/Disable
Panic Supervision*	Enabling this will have cause panic devices to send supervision signals and log a malfunction if it hasn't communicated with the panel	Enable/Disable
Temperature Units (TSP-3 only)	This changes the temperature units that are displayed on TSP-3s that are associated with the system. This does not change the units that are displayed on the customer site or mobile app.	Celsius/Fahrenheit

*Only available on certain firmware updates. For best results, please ensure your panel is on the most up to date firmware.

9.4. Appendix D: Sensor Group Configuration

Eligible Arming States:

Disarm: Lowest level of security, the panel is not armed.

Arm Stay: The system is partially armed, certain zones, such as those inside of the protected

premises, are not armed as it is expected they may be tripped while the system is Armed Stay.

Arm Away: Highest level of security, all zones that generate an alarm and report to the Central Monitoring Station are armed.

24-Hour Zones: These zones are always armed and will generate an alarm regardless of the arming state of the panel.

Sensor Groups by Physical Sensor Type:

Contact:

Group	Short Description	Long Description
8	24 Hour Burglar	24-Hour instant burglar alarm, always armed, meant to guard special belongings such as a gun cabinet or safe. Siren will sound and alarm will transmit regardless of arming state.
10	Standard Delay Entry/Exit	An Entry/Exit zone that will begin Entry Delay when tripped while the system is armed in stay or away mode. If the system is not disarmed by the end of Entry Delay, an Alarm will sound.
11	Extended Delay Entry/Exit	An Entry/Exit zone that will begin Extended Entry Delay when tripped while the system is armed. If the system is not disarmed by the end of Entry Delay, an Alarm will sound.
13	Instant Perimeter	Exterior zones, instant audible alarm during arm stay and arm away. Siren will sound.
14	Interior Stay/Away	Interior zones, instant audible alarm during arm stay and arm away. Siren will sound/
16	Interior Away	Interior zone, armed in Away mode Only, not armed in stay mode. Meant for interior doors that are used during arm stay. Siren will sound.
18	Sensor with Cross Zone Verification	Meant for zones subject to false alarms or for Monitoring Stations that require verified alarms. Tripping one sensor with cross zone verification activates the siren and reports as a trouble, while tripping a second sensor with Cross Zone verification reports a Verified alarm to the Monitoring Station. Armed in Away only
19	Interior Delay	Interior zone that starts Entry Delay when tripped. Generates an audible alarm at the end of Entry Delay if system is not disarmed. Armed in Away mode, not armed in stay mode.
25	Chime Only	No Alarm Report. Tripping sensor causes an audible chime at the panel regardless of arm state
31	Voice Chime Only	No Alarm Report. Tripping sensor causes an audible voice announcement at the panel regardless of arm state. Only eligible on specific hardware
36	24-Hour Tamper	24-Hour instant tamper alert, always armed, meant to guard special belongings such as a gun cabinet or safe. When tripped, transmits a tamper to the Monitoring Station regardless of arm

		state. Siren will not sound.
40*	Non-Reporting Zone	Used for Occupancy sensors or other zones without a Central Station Report
60	24-hours silent alarm	Used for special circumstances when a silent auxiliary panic needs to be registered using a contact regardless of arming state. Use case: to be used with a wired panic button wired into the DC-23. Alarm will sound
67	Sensor with Cross Zone verification	Meant for zones subject to false alarms or for Monitoring Stations that require verified alarms. Tripping one sensor with cross zone verification activates the siren and reports an unverified alarm, while tripping a second sensor with Cross Zone verification reports a Verified alarm to the Monitoring Station. Armed in Stay and Away.

Motion:

Group	Short Description	Long Description
13	Instant Perimeter	Exterior zones, instant audible alarm during arm stay and arm away. Siren will sound.
15	Interior Follower	Interior follower, armed in stay and away, initiates an audible alarm if an entry/exit zone (ex: front door) is not tripped first. If Entry/Exit Zone is tripped, will not generate an alarm until the end of Entry Delay. Siren will sound.
17	Interior Stay Follower	Interior follower, when armed away initiates an audible alarm if an entry/exit zone (ex: front door) is not tripped first. If Entry/Exit Zone is tripped, will not generate an alarm until the end of Entry Delay. No response during arm Stay. Siren will sound.
18	Sensor with Cross Zone Verification	Meant for zones subject to false alarms or for Monitoring Stations that require verified alarms. Tripping one sensor with cross zone verification activates the siren and reports as a trouble, while tripping a second sensor with Cross Zone verification reports a Verified alarm to the Monitoring Station. Armed in Away only
20	Interior Delay	Interior zone that starts Entry Delay when tripped. Generates an audible alarm at the end of Entry Delay if system is not disarmed. Armed in Away mode, not armed in Stay mode.
25	Motion-Chime Only	No Alarm Report. Tripping sensor causes an audible chime at the panel regardless of arm state.
31	Voice Chime Only	No Alarm Report. Tripping sensor causes an audible voice announcement at the panel regardless of arm state. Only eligible on specific hardware

40*	Non-Reporting Zone	Used for Occupancy sensors or other zones without a Central Station Report
50	Delay Stay/Away	Interior sensors, activate Entry Delay when tripped in arm stay and arm away. If the system is not disarmed by the end of Entry Delay, an Alarm will sound. Siren will sound.
67	Sensor with Cross Zone verification	Meant for zones subject to false alarms or for Monitoring Stations that require verified alarms. Tripping one sensor with cross zone verification activates the siren and reports an unverified alarm, while tripping a second sensor with Cross Zone verification reports a Verified alarm to the Monitoring Station. Armed in Stay and Away.

Glassbreak:

Group	Short Description	Long Description
8	24 Hour Burglar	24-Hour instant burglar alarm, always armed, meant to guard special belongings such as a gun cabinet or safe. Siren will sound and alarm will transmit regardless of arming state.
13	Instant Perimeter	Exterior zones, instant audible alarm during arm stay and arm away. Siren will sound.
16	Interior Away	Interior zone, armed in Away mode Only, not armed in stay mode. Meant for interior doors that are used during arm stay. Siren will sound. When triggered, this causes an instant alarm.
18	Sensor with Cross Zone Verification	Meant for zones subject to false alarms or for Monitoring Stations that require verified alarms. Tripping one sensor with cross zone verification activates the siren and reports as a trouble, while tripping a second sensor with Cross Zone verification reports a Verified alarm to the Monitoring Station. Armed in Stay and Away.
40*	Non-Reporting Zone	Used for Occupancy sensors or other zones without a Central Station Report
67	Sensor with Cross Zone verification	Meant for zones subject to false alarms or for Monitoring Stations that require verified alarms. Tripping one sensor with cross zone verification activates the siren and reports an unverified alarm, while tripping a second sensor with Cross Zone verification reports a Verified alarm to the Monitoring Station.

		Armed in Stay and Away.
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Life Safety:

Group	Short Description	Long Description
34	24-Hour Carbon Monoxide	Carbon Monoxide gas detectors, instant alarm, always armed. Generates audible CO alarm when tripped.
26	24-Hour Fire	24-Hour fire, rate-of-rise heat, and smoke sensors, instant alarm, always armed. Tripping sensor generates an audible fire alarm regardless of arming state.
61	24-Hour Fire with Verification	24-Hour fire with verification, requires a second sensor to validate the fire smoke or heat rise sensor, instant alarm, always armed. Siren will sound.
79*	24-Hour Silent Fire	24-Hour fire, rate-of-rise heat, and smoke sensors, instant alarm without siren, always armed
38	24-Hour Water	24-Hour auxiliary Water Sensors. Generates an audible Water Alarm when tripped regardless of arming state. Siren will sound and device will beep (if device has internal sounder).
91*	24-Hour Silent Freeze/Water	24 Hour Auxiliary freeze or water sensors, Generates an audible Water Alarm when tripped regardless of arming state. Alarm with no siren, device will issue trouble beeps when tripped if device has internal sounder.
40*	Non-Reporting Zone	Used for Occupancy sensors or other zones without a Central Station Report
52	24-Hour Freeze	24-Hour Auxiliary freeze sensors. Generates a Freeze Alarm when tripped regardless of arming state. Alarm with no siren, device will issue trouble beeps when tripped if device has internal sounder. When triggered, this causes an instant alarm.

Keypad:

Group	Short Description	Long Description
0	24-Hour Fixed Panic	When the panel is armed away, tampering the keypad will cause an audible panic alarm. Siren will sound.
2	24-Hour Silent Fixed Panic	When the panel is armed away, tampering the keypad will cause a silent panic alarm. Siren will not sound.
4	24-Hour Fixed Auxiliary	When the panel is armed away, tampering the keypad will cause an audible Medical panic. Siren will sound.

Keyfob:

Group	Short Description	Long Description
1	24-Hour Portable Panic	When the panic button is triggered, an audible panic alarm will occur regardless of arm state. Siren will sound.
3	24-Hour Silent Portable Panic	When the panic button is triggered, a silent panic alarm will occur regardless of arm state. Siren will not sound.
6	24-Hour Portable Auxiliary	When the panic button is triggered, an audible medical panic alarm will occur regardless of arm state. Siren will sound.

Misc.:

Group	Short Description	Long Description
72*	Security Sensor Repeater	Auxiliary: Device Repeater used to extend range
33	Siren	Wireless Siren.
30		Used to denote a device that is not recognized or supported.
40*	Non-Reporting Zone	Used for Occupancy sensors or other zones without a Central Station Report.

*Not available on certain firmware versions. For best results, please update to the latest firmware version.

9.5. Appendix E: CID Reporting Codes

CS Event Description	SIA Code	CID Code
Sensor Low Battery	XT	1384
Sensor Low Battery	XR	3384
Restore		
Sensor Tamper Alarm	TA	1383
Sensor Tamper Restore	TR	3383
AC Power Fail Restore	AR	3301
AC Power Fail	AT	1301
Closing Report	CL	3401
Closing Report	CL	3403
Recent Closing	CR	1459
Exit Alarm	EE	1457
Bus Supervisories	NULL	NULL
Fire Panic	FA	1110
Fire Panic Cancel	FH	3110
Duress Alarm	HA	1121

Touchpad Tamper - Keystroke	JA	1137
Touchpad Tamper Restore	NULL	3137
Panel Programming	LB	1627
Phone Failure Restore	LR	3350
Phone Failure	LT	1350
Auxiliary Panic	MA	1100
Auxiliary Panic Cancel	MH	3100
No Activity Alarm	NA	1102
No Activity Restore	NS	3102
Cancel Report	OC	1406
Opening Report	OP	1401
Opening Report	OP	1403
Police Panic	PA	1120
Police Panic Cancel	PH	3120
Receiver Failure	XS	1355
CPU Low Battery - Restoral	YR	3302
CPU Low Battery	YT	1302
Freeze - ESM Module	ZA	1159
Freeze - ESM Module Restore	ZH	3159
Panel Tamper	TA	1137
Manual Phone Test	RX	1601
Auto-Phone Test	RP	1602
Cellular Backup Failure	YC	1356
Cellular Backup Restore	YK	3356
Sensor Test Start	TS	1607
Sensor Test End	TE	3607
Receiver Jammed	XQ	1355
Suspected Entry Delay Alarm (Possible Crash & Smash)	UZ	1777
Two Trip Error	XM	3138
Silent Police Panic	HA	1122
Silent Police Panic Cancel	HH	3122

9.6. Appendix F: Panel Factory Reset

1. Disconnect AC adapter, slide battery switch to Off.
2. Press and hold the Learn button and connect AC adapter to the panel. All LEDs will turn on for 10 seconds.
3. Keep holding the Reset button for 30 seconds then release. After 10 seconds, the panel will restart and all LEDs will turn ON.
4. Wait for 10 seconds and all LEDs will turn OFF to indicate factory reset is complete.

9.7. Appendix G: Sensor Enrollment Instructions

Prior to enrollment, put control panel into Sensor Learn Mode. For more detailed instructions and diagrams, please see device installation manuals.

Keypad:

While in normal operation mode, press and hold * & # at the same time for 1 second. The keypad will emit a long beep to indicate it sent the enroll signal to the control panel.

Alternatively:

1. Put keypad into test mode:
2. Press *.
3. Enter the default PIN 0000.
4. Press * and then 7, the keypad will emit a long beep to indicate it sent the enroll signal to the control panel.
5. Exit test mode by pressing the disarm key twice.

Keyfob:

Press any button on the keyfob once.

Contact, PIR, Shock, Glassbreak:

Press the “Test” button on the device. If unsure which button is the “Test” button, please refer to the sensor diagram in the installation manual.

Siren:

Please refer to the siren installation guide for information on how to send the enroll signal from the siren. To put a siren into partition 2, the siren must first be enrolled into partition 1. Then, move the siren to partition 2 using AirFX or MobileTech and ensure command is acknowledged by the panel by checking the Event History. Once the command is acknowledged, follow the instructions to send the enroll signal from the siren a second time to complete the move to partition 2. If this step is not followed, the siren will not function properly.